FBISE • SSC-II

CHEMISTRY CLASS 10

SLO-based question bank

Important short questions • 3 marks | Long questions • 6 marks

120 short questions + 47 long questions

Focused model answers • Worked calculations • Labelled diagrams • Fully displayed organic structures

Matter | Stoichiometry | Electrochemistry | Kinetics | Salts | Basics of Organic Chemistry | Hydrocarbons | Hydroxy Compounds

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Based on the assessment framework supplied by the user. Only the summative theory outcomes in the eight requested content areas are used.

Practice resource, not an official FBISE paper or marking scheme. The bank covers distinct examinable skills and common question forms; no finite list can include every possible rewording or guarantee an exam question.

How to use this bank

For a 3-mark answer: give three relevant scoring points, or show the essential formula, working and final answer with units. For a 6-mark answer: give the requested explanation in a logical sequence with balanced equations, conditions, calculations or a labelled diagram as appropriate.

The numbered answer points are a suggested marking structure. Actual credit and mark allocation depend on the question and the official examiner's scheme. A six-mark question can award several marks to a diagram or calculation; six sentences alone do not guarantee six marks.

SLO references such as B-19 mean C-10-B-19. The source prints B-19 as “C-10-8-19”; its intended code is used here. Combined long questions may draw on several listed theory outcomes.

Scope decisions

• Matter: B-01 to B-07 and B-09 included. B-08 is PBA-only and excluded.

• Stoichiometry: B-11, theory portions of B-12, and B-14 included. Limiting-reactant calculations, B-10 as a standalone outcome, B-13 PBA titration calculations, and B-15 percentage yield/composition/purity are excluded.

• B-12 itself explicitly includes reacting gas volumes, so gas-volume stoichiometry is retained with the molar volume given in each question. Calculating a formula from supplied percentages under B-14 is retained; calculating percentage composition under B-15 is excluded.

• Electrochemistry: B-16 to B-27 included. Kinetics: only B-30, B-31 and B-33; formative collision theory, catalyst theory and factor explanations (B-28, B-29, B-32) are excluded.

• Salts: B-34 to B-39 included. The theoretical titration preparation method in B-39 is retained, independently of the excluded empirical titration-calculation outcome B-13.

• Organic basics E-01 to E-03; hydrocarbons E-04 to E-14; hydroxy compounds E-15 to E-19 included. No separate Carbonyl Compounds, Polymers, Biochemistry, Nitrogen/Sulfur or Oxides chapters are added.

Corrections and conventions

The source duplicates E-12. Here E-12a means petroleum separation and E-12b means fraction-property trends; these are editorial labels, not new official codes.

E-15 contains “ethane” and “6000 kPa / 6 atm”. Hydration uses ETHENE, and 6000 kPa is approximately 60 atm, not 6 atm. Corrected conditions are used throughout.

Formula conventions: -C(=O)-O- explicitly shows an ester carbonyl. Structural formulae in answers are supplemented by fully displayed formulae in the atlas. For aqueous halides, the identity of the metal, concentration and electrode material matter; “all dilute halides give oxygen” is not used as a universal rule.

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PDF bookmarks also provide direct navigation to each chapter and reference section. Question IDs are unique throughout the bank: S = short; L = long.

01 Matter

Included theory SLOs: C-10-B-01 to B-07, and B-09

Short questions | 3 marks each

S001 [3 marks] Explain why temperature stays constant while a pure solid melts at constant pressure.

SLO: C-10-B-01

1. Heat is absorbed as latent heat during melting.
2. The energy increases the potential-energy part of internal energy by overcoming attractions between particles.
3. Average kinetic energy does not increase, so temperature remains constant until melting is complete.

S002 [3 marks] Explain boiling in terms of particle energy and internal energy.

SLO: C-10-B-01

1. During boiling, energy is supplied to separate liquid particles into gas particles.
2. At the boiling point and constant pressure, absorbed latent heat increases potential energy rather than average kinetic energy.
3. Internal energy increases but temperature stays constant until the liquid has boiled away.

S003 [3 marks] Why is energy released during freezing and condensation without a temperature fall during the change?

SLO: C-10-B-01

1. Particles come closer and form stronger intermolecular attractions.
2. Their potential energy decreases and latent heat is released to the surroundings.
3. For a pure substance at constant pressure, average kinetic energy and temperature remain constant during the change.

S004 [3 marks] Distinguish sublimation from deposition using particle theory.

SLO: C-10-B-01

1. Sublimation is the direct change from solid to gas; deposition is the reverse change.
2. Sublimation absorbs energy to separate particles; deposition releases energy as attractions form.
3. Examples: iodine solid forms vapour on heating, while water vapour can deposit directly as frost.

S005 [3 marks] Differentiate kinetic energy, potential energy and internal energy of particles.

SLO: C-10-B-01

1. Kinetic energy is energy due to particle motion; average kinetic energy is related to temperature.
2. Potential energy is associated with particle positions and their mutual attractions.
3. Internal energy is the total microscopic kinetic and potential energy of the particles.

S006 [3 marks] Give three differences between evaporation and boiling.

SLO: C-10-B-02

1. Evaporation occurs only at the surface; boiling occurs throughout the liquid.
2. Evaporation can occur below the boiling point; boiling occurs at the boiling point for the prevailing pressure.
3. Evaporation is generally slow and has no bubbles within the liquid; boiling is rapid and forms vapour bubbles.

S007 [3 marks] Why does evaporation cool the remaining liquid?

SLO: C-10-B-02

1. Some faster particles at the surface have enough energy to escape.
2. They carry away energy, reducing the average kinetic energy of particles left behind.
3. The temperature of the remaining liquid falls unless energy is replaced from the surroundings.

S008 [3 marks] What do a sloping line and a horizontal line on a heating curve show?

SLO: C-10-B-03

1. A sloping line shows heating within one state, with rising temperature and average kinetic energy.
2. A horizontal line shows a change of state at constant temperature for a pure substance at constant pressure.
3. Energy absorbed along a horizontal line changes particle separation and potential energy.

S009 [3 marks] Interpret the two plateaus of the cooling curve of a pure substance.

SLO: C-10-B-03

1. The higher-temperature plateau represents condensation of gas into liquid.
2. The lower-temperature plateau represents freezing of liquid into solid.
3. At each plateau latent heat is released while temperature remains constant until the change is complete.

S010 [3 marks] State Boyle's law and explain it using particle theory.

SLO: C-10-B-04

1. For a fixed amount of gas at constant temperature, pressure is inversely proportional to volume: $PV =$ constant.
2. Reducing volume gives particles less space, so they collide more frequently with the container walls.
3. The increased collision frequency per unit area increases pressure; average kinetic energy is unchanged.

S011 [3 marks] State Charles' law and explain it using particle theory.

SLO: C-10-B-04

1. For a fixed amount of gas at constant pressure, volume is proportional to absolute temperature: $V/T =$ constant.
2. Heating increases average particle kinetic energy and speed.
3. The gas expands so that pressure remains constant; temperature must be measured in kelvin.

S012 [3 marks] State Avogadro's law and give its particle explanation.

SLO: C-10-B-04

1. At constant temperature and pressure, gas volume is proportional to the number of moles: V/n = constant.
2. Adding particles would increase collisions with the walls if volume stayed fixed.
3. Volume therefore increases in the same proportion to maintain pressure; equal gas volumes at the same T and P contain equal numbers of molecules.

S013 [3 marks] A sealed rigid gas container is heated. Explain the change in pressure.

SLO: C-10-B-04

1. Gas particles gain average kinetic energy and move faster.
2. They strike the walls more frequently and with greater momentum changes.
3. Pressure increases because the container volume and amount of gas are fixed.

S014 [3 marks] Explain why water boils at a lower temperature on a mountain.

SLO: C-10-B-05

1. Atmospheric pressure is lower at high altitude.
2. Boiling begins when the liquid vapour pressure equals the external pressure.
3. This equality is reached at a lower temperature, so the boiling point decreases.

S015 [3 marks] How does external pressure affect boiling and evaporation?

SLO: C-10-B-05

1. Increasing external pressure raises the boiling point; decreasing it lowers the boiling point.
2. At fixed temperature, sufficiently low pressure can make a liquid boil because vapour bubbles can expand.
3. Lower gas pressure generally makes escape of vapour easier and can increase net evaporation; evaporation also depends on airflow and vapour already present.

S016 [3 marks] Explain gas diffusion with an example.

SLO: C-10-B-06

1. Diffusion is the net spreading of particles from a region of higher concentration to one of lower concentration.
2. Gas particles move continuously and randomly, spreading through available space.
3. For example, perfume vapour spreads through a room until it is more evenly distributed.

S017 [3 marks] How do molecular mass and temperature affect gas diffusion?

SLO: C-10-B-07

1. At the same temperature, lighter gas molecules generally diffuse faster than heavier ones.
2. Lighter particles have greater average speeds at the same average kinetic energy.
3. Raising temperature increases average kinetic energy and speed, so diffusion becomes faster.

S018 [3 marks] Why is the diffusion rate of a medicine important in the body?

SLO: C-10-B-09

1. Medicine molecules must spread from a region of higher concentration to a region of lower concentration in body fluids or tissues.
2. An adequate diffusion rate helps the drug reach its target in sufficient concentration and time.
3. Very slow diffusion can delay its action; delivery systems can control release to sustain useful concentrations. Diffusion is one part of transport, alongside blood circulation.

Long questions | 6 marks each

L001 [6 marks] Describe all stages of a heating curve of a pure substance using particle theory.

SLO: C-10-B-01, B-03

1. Solid warming: temperature rises as particles vibrate more strongly about fixed positions.
2. Melting plateau: solid and liquid coexist; latent heat increases potential energy, while temperature remains constant.
3. Liquid warming: temperature rises as average kinetic energy increases.
4. Boiling plateau: liquid and gas coexist; latent heat separates particles without increasing average kinetic energy.
5. Gas warming: once boiling is complete, further heating raises gas temperature and average kinetic energy.
6. The reverse stages occur during cooling; condensation and freezing release latent heat. See the labelled heating-curve diagram.

L002 [6 marks] Compare evaporation with boiling and explain the effect of external pressure.

SLO: C-10-B-01, B-02, B-05

1. Evaporation is a surface process; boiling occurs throughout a liquid.
2. Evaporation can take place below the boiling point; boiling takes place when vapour pressure equals external pressure.
3. Evaporation has no vapour bubbles within the bulk liquid; boiling produces bubbles throughout.
4. Evaporation preferentially removes high-energy particles and can cool the liquid.
5. Higher external pressure raises the boiling point; lower external pressure lowers it.
6. For example, a pressure cooker raises the boiling temperature, whereas water boils at a lower temperature on mountains. Lower ambient gas pressure generally assists net evaporation under otherwise comparable conditions.

L003 [6 marks] Explain Boyle's, Charles' and Avogadro's laws with their fixed conditions and particle interpretations.

SLO: C-10-B-04

1. Boyle: for fixed n and T, P is inversely proportional to V; $P_1V_1 = P_2V_2$.
2. At constant temperature, compression increases collision frequency at the walls and therefore pressure.
3. Charles: for fixed n and P, V is proportional to T in kelvin; $V_1/T_1 = V_2/T_2$.
4. Heating speeds up particles, so expansion is needed to keep pressure constant.
5. Avogadro: at fixed T and P, V is proportional to n; $V_1/n1 = V_2/n2$.
6. Adding more particles requires proportionally more space to maintain the same pressure. Do not use Celsius in Charles' law.

L004 [6 marks] Explain diffusion, factors affecting it and its importance in drug delivery.

SLO: C-10-B-06, B-07, B-09

1. Gas diffusion results from continuous random particle movement.
2. The net spreading is from higher to lower concentration until a more uniform distribution is reached.
3. At the same temperature, lower-molecular-mass gases generally diffuse faster because their average speed is higher.
4. Increasing temperature raises average particle speed and generally speeds diffusion.
5. In the body, medicine molecules diffuse through fluids and suitable membranes toward lower-concentration regions.
6. The rate affects how quickly an effective concentration reaches a target; controlled release can maintain action over time. Blood circulation also transports drugs.

L005 [6 marks] A substance warms from -10°C to 0°C, stays at 0°C, warms to 100°C, stays at 100°C, then warms further. Interpret the five stages at 1 atm.

SLO: C-10-B-01, B-03

1. From -10°C to 0°C, ice warms: its average kinetic energy increases.
2. At 0°C, ice melts: supplied energy increases potential energy, not temperature.
3. Between 0°C and 100°C, liquid water warms: average kinetic energy rises.
4. At 100°C, water boils: latent heat separates particles into vapour.
5. After all water vaporises, steam warms above 100°C and its average kinetic energy rises.
6. The plateaus correspond to two states coexisting; the stated temperatures apply to pure water at approximately 1 atm.

02 Stoichiometry

Included theory SLOs: C-10-B-11; summative portions of B-12; B-14

Short questions | 3 marks each

S019 [3 marks] Define concentration and distinguish g dm^{-3} from mol dm^{-3}.

SLO: C-10-B-11

1. Concentration is the quantity of solute present per unit volume of solution.
2. Mass concentration = mass of solute in g / volume of solution in dm^3; unit g dm^{-3}.
3. Molar concentration = amount of solute in mol / volume of solution in dm^3; unit mol dm^{-3}.

S020 [3 marks] Convert 5.85 g dm^{-3} NaCl into mol dm^{-3}. Use Mr(NaCl) = 58.5.

SLO: C-10-B-11

1. Use c(mol dm^{-3}) = mass concentration / molar mass.
2. c = 5.85 / 58.5.
3. Concentration = 0.100 mol dm^{-3}.

S021 [3 marks] Find the mass concentration of 0.200 mol dm^{-3} NaOH. Use Mr(NaOH) = 40.

SLO: C-10-B-11

1. Mass concentration = molar concentration × molar mass.
2. Mass concentration = 0.200 × 40.
3. Answer = 8.00 g dm^{-3}.

S022 [3 marks] Calculate the concentration when 4.00 g NaOH is made up to 250 cm^3 of solution. Mr = 40.

SLO: C-10-B-11, B-12

1. Volume = 250/1000 = 0.250 dm^3.
2. Mass concentration = 4.00/0.250 = 16.0 g dm^{-3}.
3. Molar concentration = 16.0/40 = 0.400 mol dm^{-3}.

S023 [3 marks] Calculate the mass of NaCl needed to prepare 200 cm^3 of 0.150 mol dm^{-3} solution. Mr = 58.5.

SLO: C-10-B-12

1. V = 200/1000 = 0.200 dm^3.
2. n = cV = 0.150 × 0.200 = 0.0300 mol.
3. m = nM = 0.0300 × 58.5 = 1.755 g, approximately 1.76 g.

S024 [3 marks] Calculate the mass of MgO formed from 4.80 g Mg in excess oxygen. Ar: Mg = 24, O = 16.

SLO: C-10-B-12

1. $2Mg + O_2 \rightarrow 2MgO$; $n(Mg) = 4.80/24 = 0.200$ mol.
2. The Mg:MgO ratio is 1:1, so $n(MgO) = 0.200$ mol.
3. Mass of MgO = $0.200 \times 40 = 8.00$ g.

S025 [3 marks] Calculate the mass of oxygen required to burn 3.20 g methane completely. Ar: C = 12, H = 1, O = 16.

SLO: C-10-B-12

1. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$; $n(CH_4) = 3.20/16 = 0.200$ mol.
2. $n(O_2) = 2 \times 0.200 = 0.400$ mol.
3. Mass of $O_2 = 0.400 \times 32 = 12.8$ g.

S026 [3 marks] What mass of $CaCO_3$ produces 4.80 dm^3 CO_2 at r.t.p. on complete reaction with excess acid? Molar gas volume = 24.0 dm^3 mol^{-1}; Mr($CaCO_3$) = 100.

SLO: C-10-B-12

1. $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$; $n(CO_2) = 4.80/24.0 = 0.200$ mol.
2. The $CaCO_3$:CO_2 ratio is 1:1, so $n(CaCO_3) = 0.200$ mol.
3. Mass of $CaCO_3 = 0.200 \times 100 = 20.0$ g.

S027 [3 marks] Calculate the volume of CO_2 produced by complete combustion of 50 cm^3 ethene with excess oxygen at the same temperature and pressure.

SLO: C-10-B-12

1. $C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$.
2. At the same temperature and pressure, gaseous volumes follow the mole ratio: ethene:CO_2 = 1:2.
3. Volume of $CO_2 = 2 \times 50 = 100$ cm^3.

S028 [3 marks] What volume of 0.100 mol dm^{-3} HCl reacts with 25.0 cm^3 of 0.200 mol dm^{-3} NaOH?

SLO: C-10-B-12

1. $HCl + NaOH \rightarrow NaCl + H_2O$; $n(NaOH) = 0.200 \times 0.0250 = 0.00500$ mol.
2. Ratio HCl:NaOH = 1:1, so $n(HCl) = 0.00500$ mol.
3. $V(HCl) = n/c = 0.00500/0.100 = 0.0500$ dm^3 = 50.0 cm^3.

S029 [3 marks] Differentiate empirical and molecular formulae with an example.

SLO: C-10-B-14

1. An empirical formula gives the simplest whole-number ratio of atoms of each element.
2. A molecular formula gives the actual number of atoms of each element in a molecule.
3. Glucose has empirical formula CH_2O and molecular formula $C_6H_{12}O_6$; the latter is six times the empirical formula.

S030 [3 marks] Find the empirical formula of a compound containing 24 g carbon and 6 g hydrogen. Ar: C = 12, H = 1.

SLO: C-10-B-14

1. Amounts: C = 24/12 = 2 mol; H = 6/1 = 6 mol.
2. Divide by the smaller value: C:H = 2/2:6/2 = 1:3.
3. Empirical formula = CH_3.

S031 [3 marks] A compound has empirical formula CH_2O and Mr = 180. Find its molecular formula.

SLO: C-10-B-14

1. Empirical formula mass = 12 + 2(1) + 16 = 30.
2. Multiplier = molecular mass / empirical formula mass = 180/30 = 6.
3. Molecular formula = $(CH_2O)_6 = C_6H_{12}O_6$.

S032 [3 marks] Find the empirical formula of a compound containing 40.0% C, 6.7% H and 53.3% O. Ar: C = 12, H = 1, O = 16.

SLO: C-10-B-14

1. For 100 g, amounts are C = 40.0/12 = 3.33; H = 6.7/1 = 6.7; O = 53.3/16 = 3.33 mol.
2. Divide each by 3.33: C:H:O ≈ 1:2:1.
3. Empirical formula = CH_2O.

Long questions | 6 marks each

L006 [6 marks] A student needs 500 cm^3 of 0.250 mol dm^{-3} sodium carbonate solution. Calculate the required mass and its concentration in g dm^{-3}. Mr(Na_2CO_3) = 106.

SLO: C-10-B-11, B-12

1. Convert volume: 500 cm^3 = 0.500 dm^3.
2. Use n = cV.
3. n(Na_2CO_3) = 0.250 × 0.500 = 0.125 mol.
4. Use m = nM: m = 0.125 × 106 = 13.25 g.
5. Mass concentration = cM = 0.250 × 106 = 26.5 g dm^{-3}.
6. Therefore dissolve 13.25 g and make up to a final solution volume of 500 cm^3; do not simply add 500 cm^3 water.

L007 [6 marks] 10.0 g $CaCO_3$ reacts completely with excess hydrochloric acid. Calculate the mass of $CaCl_2$ and volume of CO_2 at r.t.p. Ar: Ca = 40, C = 12, O = 16, Cl = 35.5. Use 24.0 dm^3 mol^{-1}.

SLO: C-10-B-12

1. Balanced equation: $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$.
2. M($CaCO_3$) = 100 g mol^{-1}; n = 10.0/100 = 0.100 mol.
3. Mole ratios $CaCO_3$:$CaCl_2$:CO_2 are 1:1:1.
4. n($CaCl_2$) = 0.100 mol; M($CaCl_2$) = 111 g mol^{-1}.
5. Mass $CaCl_2$ = 0.100 × 111 = 11.1 g.
6. n(CO_2) = 0.100 mol; V = 0.100 × 24.0 = 2.40 dm^3 at r.t.p.

L008 [6 marks] Calculate the volume of 0.200 mol dm^{-3} NaOH needed to neutralise 25.0 cm^3 of 0.100 mol dm^{-3} H_2SO_4. Also calculate the mass of Na_2SO_4 formed. Mr(Na_2SO_4) = 142.

SLO: C-10-B-12

1. $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$.
2. V(acid) = 25.0/1000 = 0.0250 dm^3.
3. n(H_2SO_4) = cV = 0.100 × 0.0250 = 0.00250 mol.
4. n(NaOH) = 2 × 0.00250 = 0.00500 mol.
5. V(NaOH) = 0.00500/0.200 = 0.0250 dm^3 = 25.0 cm^3.
6. n(Na_2SO_4) = 0.00250 mol; mass = 0.00250 × 142 = 0.355 g. This is equation-based stoichiometry, not an empirical titration-data exercise.

L009 [6 marks] A compound contains 54.5% C, 9.1% H and 36.4% O. Its Mr is 88. Determine its empirical and molecular formulae. Ar: C = 12, H = 1, O = 16.

SLO: C-10-B-14

1. Assume 100 g: C = 54.5 g, H = 9.1 g, O = 36.4 g.
2. Divide by atomic masses: 54.5/12 = 4.54; 9.1/1 = 9.10; 36.4/16 = 2.275 mol.
3. Divide by 2.275 to get approximately 2:4:1.
4. Empirical formula = C_2H_4O; empirical formula mass = 24 + 4 + 16 = 44.
5. Multiplier = 88/44 = 2.
6. Molecular formula = $C_4H_8O_2$. Show the ratio calculation as well as both final formulae.

L010 [6 marks] An oxide contains 5.60 g iron and 2.40 g oxygen. Determine its empirical formula and explain how fractional ratios are handled. Ar: Fe = 56, O = 16.

SLO: C-10-B-14

1. n(Fe) = 5.60/56 = 0.100 mol.
2. n(O) = 2.40/16 = 0.150 mol.
3. Divide by 0.100 to give Fe:O = 1:1.5.
4. Multiply both ratios by 2 to obtain whole numbers 2:3.
5. The empirical formula is Fe_2O_3.
6. Do not round 1.5 to 2; multiply the entire ratio by the smallest suitable whole number.

L011 [6 marks] For complete combustion of 2.40 dm^3 ethene, find the oxygen volume required and the CO_2 volume produced at r.t.p. Also find the mass of CO_2. Use 24.0 dm^3 mol^{-1} and Mr(CO_2) = 44.

SLO: C-10-B-12

1. $C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$.
2. At the same T and P, gaseous volume ratio C_2H_4:O_2:CO_2 = 1:3:2.
3. Oxygen volume = 3 × 2.40 = 7.20 dm^3.
4. Carbon dioxide volume = 2 × 2.40 = 4.80 dm^3.
5. n(CO_2) = 4.80/24.0 = 0.200 mol.
6. Mass of CO_2 = 0.200 × 44 = 8.80 g. Do not include liquid water in gas-volume ratios at r.t.p.

03 Electrochemistry

Included theory SLOs: C-10-B-16 to B-27

Short questions | 3 marks each

S033 [3 marks] Define electrolysis and explain why a solid ionic compound normally cannot be electrolysed.

SLO: C-10-B-16

1. Electrolysis is decomposition of an ionic compound, molten or in aqueous solution, by passage of electric current.
2. The electrolyte must contain mobile ions to carry charge.
3. In the solid, ions are fixed in a lattice and cannot migrate to the electrodes.

S034 [3 marks] State the signs of electrodes and direction of charge movement in an electrolytic cell.

SLO: C-10-B-17, B-18

1. The anode is positive; anions move to it and lose electrons (oxidation).
2. The cathode is negative; cations move to it and gain electrons (reduction).
3. Electrons move through external wires: from the negative supply terminal to the cathode, and from the anode toward the positive supply terminal.

S035 [3 marks] How is charge carried through the wires and through the electrolyte?

SLO: C-10-B-18

1. Mobile electrons carry charge through metal wires.
2. Mobile positive and negative ions carry charge through the electrolyte, moving in opposite directions.
3. Electrons are transferred at the electrode surfaces; they do not flow through the solution as they do through a wire.

S036 [3 marks] Give the electrode products and half-equations for molten lead(II) chloride.

SLO: C-10-B-19, B-24

1. Cathode: $Pb^{2+} + 2e^- \rightarrow Pb$; molten lead forms silvery droplets.
2. Anode: $2Cl^- \rightarrow Cl_2 + 2e^-$; chlorine gas bubbles form.
3. Overall: $PbCl_2(l) \rightarrow Pb(l) + Cl_2(g)$, using inert electrodes.

S037 [3 marks] State the products and observations when concentrated brine is electrolysed using inert electrodes.

SLO: C-10-B-19

1. At the cathode, colourless hydrogen bubbles form; hydrogen gives a squeaky pop with a lighted splint.
2. At the anode, chlorine forms; it bleaches damp litmus paper after initially acidifying it.
3. Sodium hydroxide remains in solution; sodium metal is not deposited from aqueous brine.

S038 [3 marks] Write half-equations and the overall reaction for electrolysis of concentrated brine.

SLO: C-10-B-19, B-24

1. Cathode: $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$.
2. Anode: $2Cl^- \rightarrow Cl_2 + 2e^-$.
3. Overall: $2NaCl + 2H_2O \rightarrow 2NaOH + H_2 + Cl_2$; products must be kept separate in manufacture.

S039 [3 marks] Describe the products of electrolysis of dilute sulfuric acid with inert electrodes.

SLO: C-10-B-19, B-24

1. Cathode: $2H^+ + 2e^- \rightarrow H_2$; colourless hydrogen bubbles are produced.
2. Anode: $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$; colourless oxygen bubbles are produced.
3. Overall: $2H_2O \rightarrow 2H_2 + O_2$; hydrogen and oxygen volumes are in a 2:1 ratio at the same T and P.

S040 [3 marks] What happens when dilute $CuSO_4$ is electrolysed with inert electrodes?

SLO: C-10-B-22, B-24

1. Cathode: $Cu^{2+} + 2e^- \rightarrow Cu$; a reddish-brown copper coating appears.
2. Anode: $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$; oxygen bubbles form.
3. The blue colour gradually fades as Cu^{2+} is removed, and the solution becomes more acidic.

S041 [3 marks] What happens when $CuSO_4$ is electrolysed using copper electrodes?

SLO: C-10-B-22, B-24

1. Cathode: $Cu^{2+} + 2e^- \rightarrow Cu$; the cathode gains mass.
2. Anode: $Cu \rightarrow Cu^{2+} + 2e^-$; the anode loses mass.
3. Copper is transferred from anode to cathode; ideally the Cu^{2+} concentration and blue colour remain approximately constant.

S042 [3 marks] Predict the products of molten NaCl and aqueous concentrated NaCl with inert electrodes.

SLO: C-10-B-23

1. Molten NaCl gives sodium at the cathode and chlorine at the anode because only Na^+ and Cl^- are present.
2. Concentrated aqueous NaCl gives hydrogen at the cathode and chlorine at the anode.
3. In water, water is reduced in preference to Na^+; Na^+ remains in solution with OH^-.

S043 [3 marks] How can dilution change the anode product in aqueous sodium chloride electrolysis?

SLO: C-10-B-23

1. Concentrated aqueous NaCl normally gives predominantly chlorine at an inert anode.
2. Very dilute aqueous NaCl normally gives predominantly oxygen at an inert anode.
3. The discharge depends on competing oxidation processes and ion concentrations; this is the usual school-level prediction, and mixed gases can occur under some conditions.

S044 [3 marks] Predict the products of concentrated aqueous $CuBr_2$ electrolysis with inert electrodes.

SLO: C-10-B-23, B-24

1. Copper forms at the cathode: $Cu^{2+} + 2e^- \rightarrow Cu$.
2. Bromine forms at the anode: $2Br^- \rightarrow Br_2 + 2e^-$.
3. A reddish-brown copper coating and orange-brown bromine near the anode are expected; the copper-ion colour decreases.

S045 [3 marks] What checks are needed when writing ionic half-equations?

SLO: C-10-B-24

1. Write the correct species and product at the specified electrode.
2. Balance atoms, then balance total charge by adding electrons.
3. Electrons appear on the left for cathode reduction and on the right for anode oxidation; for example $Cu^{2+} + 2e^- \rightarrow Cu$.

S046 [3 marks] Describe the essential arrangement for copper electroplating an iron spoon.

SLO: C-10-B-25

1. Clean the iron spoon and connect it as the negative cathode.
2. Use copper as the positive anode and copper sulfate solution as the electrolyte.
3. Pass direct current: Cu^{2+} gains electrons on the spoon, while the copper anode dissolves to replenish Cu^{2+}.

S047 [3 marks] Give three applications of electroplating.

SLO: C-10-B-25

1. Chromium plating gives suitable metal objects a hard, attractive and corrosion-resistant surface.
2. Silver plating coats cutlery or ornaments with a thin decorative silver layer.
3. Gold plating protects electrical contacts from corrosion while using much less gold than a solid-gold part.

S048 [3 marks] Identify the electrodes, electron flow and role of the salt bridge in a Daniell cell.

SLO: C-10-B-26

1. Zinc is the negative anode and is oxidised: $Zn \rightarrow Zn^{2+} + 2e^-$.
2. Copper is the positive cathode: $Cu^{2+} + 2e^- \rightarrow Cu$; electrons flow from zinc to copper through the external circuit.
3. The salt bridge allows ion movement to maintain electrical neutrality and complete the internal circuit.

S049 [3 marks] Give three differences between electrolytic and voltaic cells.

SLO: C-10-B-26

1. An electrolytic cell uses electrical energy to drive a chemical reaction; a voltaic cell converts a spontaneous chemical reaction into electrical energy.
2. The electrolytic anode is positive; the voltaic anode is negative.
3. The electrolytic cathode is negative; the voltaic cathode is positive. Oxidation is at the anode and reduction at the cathode in both.

S050 [3 marks] A Zn-Cu cell gives 1.10 V with zinc as the negative electrode. Which metal is more reactive? Explain.

SLO: C-10-B-27

1. The negative electrode of this working voltaic cell is the anode, where oxidation occurs.
2. Zinc supplies electrons by $Zn \rightarrow Zn^{2+} + 2e^-$; copper ions gain those electrons.
3. Zinc is therefore more readily oxidised and more reactive than copper. The magnitude 1.10 V alone would not identify the more reactive metal without polarity information.

S051 [3 marks] State the reactants, energy conversion and product of a hydrogen-oxygen fuel cell.

SLO: C-10-B-20

1. Hydrogen and oxygen are continuously supplied as reactants.
2. Their chemical energy is converted directly into electrical energy.
3. Water is the only chemical product: $2H_2 + O_2 \rightarrow 2H_2O$.

S052 [3 marks] Give advantages and a limitation of hydrogen-oxygen fuel cells in vehicles.

SLO: C-10-B-21

1. They produce water at the point of use, avoiding carbon dioxide emissions from the cell itself.
2. They can convert chemical energy to electricity efficiently and operate quietly.
3. Hydrogen storage and supply are difficult and costly; producing hydrogen may use fossil energy and cause emissions elsewhere.

S053 [3 marks] Predict the products of aqueous potassium iodide electrolysis with inert electrodes.

SLO: C-10-B-23, B-24

1. Hydrogen forms at the cathode: $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$.
2. Iodide is oxidised at the anode: $2I^- \rightarrow I_2 + 2e^-$.
3. A brown iodine-containing solution forms near the anode; starch gives a blue-black colour with iodine. Potassium is not deposited from water.

S054 [3 marks] Predict the products of concentrated aqueous potassium bromide electrolysis using inert electrodes.

SLO: C-10-B-23, B-24

1. At the cathode, water forms hydrogen: $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$.
2. At the anode, bromide forms bromine: $2Br^- \rightarrow Br_2 + 2e^-$.
3. Expect hydrogen bubbles and an orange-brown bromine colour near the anode; K^+ stays in solution.

S055 [3 marks] Why is the rule “all dilute halide solutions give oxygen at the anode” unreliable?

SLO: C-10-B-23

1. Anode discharge depends on the particular halide, concentration and electrode conditions.
2. Very dilute chloride normally favours oxygen, whereas bromide and iodide can still be oxidised to their halogens.
3. Aqueous fluoride normally gives oxygen at an inert anode, not fluorine; identify the ion before applying a prediction rule.

Long questions | 6 marks each

L012 [6 marks] Describe an electrolytic cell and explain the electrolysis of molten lead(II) chloride with equations and observations.

SLO: C-10-B-16 to B-19, B-24

1. Use a direct-current source and two inert electrodes immersed in molten $PbCl_2$; the labelled general cell is shown in the diagram section.
2. The positive electrode is the anode; the negative electrode is the cathode.
3. Molten $PbCl_2$ contains mobile Pb^{2+} and Cl^- ions; Pb^{2+} moves to the cathode and Cl^- to the anode.
4. Cathode reduction: $Pb^{2+} + 2e^- \rightarrow Pb$; silvery liquid lead droplets form.
5. Anode oxidation: $2Cl^- \rightarrow Cl_2 + 2e^-$; chlorine bubbles form.
6. Overall: $PbCl_2(l) \rightarrow Pb(l) + Cl_2(g)$. Electrons travel through wires; ions carry charge through the melt.

L013 [6 marks] Explain the electrolysis of concentrated aqueous sodium chloride using inert electrodes.

SLO: C-10-B-19, B-23, B-24

1. The electrolyte contains Na^+ and Cl^-, and water provides possible competing electrode reactions.
2. At the negative cathode, water is reduced: $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$.
3. At the positive anode, chloride is oxidised: $2Cl^- \rightarrow Cl_2 + 2e^-$.
4. Hydrogen gives a squeaky pop with a lighted splint; chlorine bleaches damp litmus paper.
5. Na^+ and OH^- remain, producing sodium hydroxide solution; overall $2NaCl + 2H_2O \rightarrow 2NaOH + H_2 + Cl_2$.
6. Sodium metal is not deposited from water. In very dilute NaCl, oxygen is normally the main anode product instead; concentrated brine favours chlorine.

L014 [6 marks] Describe electrolysis of dilute sulfuric acid, including equations, observations and gas identification.

SLO: C-10-B-19, B-24

1. Add dilute sulfuric acid to water to provide ions and use inert electrodes connected to direct current.
2. Hydrogen forms at the cathode: $2H^+ + 2e^- \rightarrow H_2$.
3. Oxygen forms at the anode: $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$.
4. Double the cathode equation and add: $2H_2O \rightarrow 2H_2 + O_2$.
5. Both gases are colourless; at equal T and P, the hydrogen volume is twice the oxygen volume.
6. Hydrogen gives a squeaky pop with a lighted splint; oxygen relights a glowing splint. Sulfuric acid mainly provides conductivity and is not consumed overall.

L015 [6 marks] Compare electrolysis of dilute $CuSO_4$ with inert electrodes and with copper electrodes.

SLO: C-10-B-22, B-24

1. In both arrangements, cathode reduction is $Cu^{2+} + 2e^- \rightarrow Cu$; reddish-brown copper is deposited.
2. With inert electrodes, water is oxidised at the anode: $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$.
3. Oxygen bubbles form; Cu^{2+} concentration falls, the blue colour fades, and acidity increases.
4. With copper electrodes, the anode dissolves: $Cu \rightarrow Cu^{2+} + 2e^-$.
5. Ideally the copper-anode mass loss equals the copper-cathode mass gain; no oxygen is expected at the dissolving copper anode.
6. The dissolved copper replaces removed Cu^{2+}, so solution colour and concentration remain approximately constant. Electrode material changes the anode reaction.

L016 [6 marks] Explain copper electroplating of an iron object, with a labelled arrangement, equations and uses.

SLO: C-10-B-25, B-24

1. Remove grease and oxide from the iron object so the coating adheres properly.
2. Connect the object to the negative terminal as cathode and a copper sheet to the positive terminal as anode.
3. Immerse both in copper sulfate electrolyte without letting them touch; use the electrolytic-cell diagram with these labels.
4. At the cathode: $Cu^{2+} + 2e^- \rightarrow Cu$; copper coats the iron.
5. At the anode: $Cu \rightarrow Cu^{2+} + 2e^-$; copper dissolves and replenishes ions.
6. Electroplating provides decorative or protective surface coatings, for example silver-plated cutlery or chromium-plated fittings. An intact coating is important; exposed iron under copper can corrode.

L017 [6 marks] Draw and explain the operation of a Daniell cell.

SLO: C-10-B-26

1. The labelled diagram shows a zinc strip in $ZnSO_4$ and a copper strip in $CuSO_4$, joined by an external wire and a salt bridge.
2. Zinc is the negative anode: $Zn \rightarrow Zn^{2+} + 2e^-$.
3. Copper is the positive cathode: $Cu^{2+} + 2e^- \rightarrow Cu$.
4. Electrons flow externally from zinc to copper; zinc loses mass and copper gains mass.
5. The salt bridge completes the ionic circuit: anions move toward the zinc side and cations toward the copper side to balance charge.
6. Overall: $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$. Chemical energy is converted into electrical energy; no external power supply drives the cell.

L018 [6 marks] Three cells with common metal M give: X-M = 1.50 V (X negative), Y-M = 0.40 V (Y negative), and Z-M = 0.30 V (M negative). Rank X, Y, M and Z in reactivity under comparable conditions.

SLO: C-10-B-27

1. A negative electrode in a spontaneous voltaic cell is oxidised more readily than the positive one.
2. X is more reactive than M because X is negative in X-M.
3. Y is more reactive than M because Y is negative in Y-M.
4. With the same reference M and comparable ion conditions, X is more readily oxidised than Y because its cell voltage relative to M is larger.
5. M is more reactive than Z because M is negative in M-Z.
6. Thus X > Y > M > Z. Use polarity and a common reference; unrelated voltage magnitudes alone are insufficient.

L019 [6 marks] Describe a hydrogen-oxygen fuel cell and compare it with a petrol engine for vehicles.

SLO: C-10-B-20, B-21

1. A fuel cell receives a continuing supply of hydrogen and oxygen and converts chemical energy into electricity.
2. Overall reaction: $2H_2 + O_2 \rightarrow 2H_2O$; water is its only chemical product.
3. Unlike petrol combustion, the cell itself produces no carbon dioxide, soot or carbon monoxide.
4. Fuel cells can be efficient and quiet, with electricity driving an electric motor.
5. Hydrogen is difficult to store and transport because it requires specialised tanks or other storage methods; infrastructure and cell materials are costly.
6. Hydrogen manufacture may consume fossil fuels, so total environmental benefit depends on how hydrogen and energy are produced. Petrol has more established refuelling infrastructure.

L020 [6 marks] Predict aqueous-halide electrolysis products at inert electrodes for concentrated NaCl, very dilute NaCl, concentrated KBr and KI; give relevant half-equations.

SLO: C-10-B-23, B-24

1. For the listed alkali-metal salt solutions, hydrogen forms at the cathode: $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$.
2. Concentrated NaCl normally gives chlorine: $2Cl^- \rightarrow Cl_2 + 2e^-$.
3. Very dilute NaCl normally gives mainly oxygen: $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$.
4. Concentrated KBr gives bromine: $2Br^- \rightarrow Br_2 + 2e^-$.
5. Aqueous KI gives iodine: $2I^- \rightarrow I_2 + 2e^-$.
6. The alkali-metal ions remain in solution. Concentration and electrode conditions affect competition, so oxygen is not a universal product of every dilute halide solution.

04 Kinetics

Included theory SLOs: C-10-B-30, B-31 and B-33 only

Focus: measurement, data/graph interpretation and food-industry applications. Formative theory topics are intentionally absent.

Short questions | 3 marks each

S056 [3 marks] Define reaction rate and name two measurable changes used to follow it.

SLO: C-10-B-30

1. Reaction rate is the change in quantity of reactant consumed or product formed per unit time.
2. A gas-producing reaction can be followed by gas volume increasing with time.
3. It may also be followed by decreasing mass when gas escapes, or by a suitable temperature change measured over time.

S057 [3 marks] How can gas formation be used to measure reaction rate?

SLO: C-10-B-30

1. Collect the gas in a gas syringe and record its volume at known time intervals.
2. Average rate = increase in gas volume / time taken, for example in $cm^3\ s^{-1}$.
3. A larger volume increase in the same time indicates a faster average rate; use a leak-tight apparatus.

S058 [3 marks] Why can the mass of a reacting flask decrease, and how is rate calculated?

SLO: C-10-B-30

1. If a gaseous product escapes from the weighed flask, its measured mass decreases.
2. Record mass at regular times and calculate mass lost.
3. Average rate = mass lost / elapsed time, usually in $g\ s^{-1}$; this does not mean total mass is destroyed.

S059 [3 marks] How can temperature change help monitor a reaction?

SLO: C-10-B-30

1. An exothermic reaction may raise the mixture temperature; an endothermic reaction may lower it.
2. Record temperature at regular intervals under comparable, well-insulated conditions.
3. A faster temperature change can indicate faster reaction, but heat loss and heat capacity affect the result, so temperature change is not a universal direct measure of chemical amount.

S060 [3 marks] A reaction produces 36 cm^3 gas in 60 s. Calculate its average rate and explain what the value means.

SLO: C-10-B-31

1. Average rate = gas volume formed / time.
2. Rate = 36/60 = 0.60 $cm^3\ s^{-1}$.
3. The average gas production is 0.60 cm^3 each second over this interval; the instantaneous rate need not remain constant.

S061 [3 marks] What do slope and a plateau mean on a gas-volume versus time graph?

SLO: C-10-B-31

1. The slope represents the rate of gas production.
2. A steeper slope indicates a faster rate; a progressively flatter curve indicates a decreasing rate.
3. A horizontal plateau shows no further gas is being collected: the reaction has effectively stopped or the gas-producing reactant is used up.

S062 [3 marks] Two curves reach the same final gas volume, but A reaches it sooner. Compare the reactions.

SLO: C-10-B-31

1. A has the greater average rate because it produces the same gas volume in less time.
2. If A has a steeper initial slope, its initial rate is also greater.
3. Equal final gas volumes at the same T and P mean equal amounts of gas, not equal reaction rates.

S063 [3 marks] A flask mass falls from 120.00 g to 119.40 g in 30 s. Calculate the average mass-loss rate.

SLO: C-10-B-31

1. Mass loss = 120.00 - 119.40 = 0.60 g.
2. Average mass-loss rate = 0.60/30.
3. Answer = 0.020 g s^{-1}; the gas lost from the flask accounts for the lower measured mass.

S064 [3 marks] Why is chemical kinetics important when choosing harvest and transport times for fruit?

SLO: C-10-B-33

1. Kinetics helps estimate how quickly ripening and spoilage changes occur.
2. Harvest time can be chosen so produce reaches suitable ripeness when it arrives at market.
3. Storage and transport conditions and journey length can then be planned to preserve quality and reduce waste.

Long questions | 6 marks each

L021 [6 marks] Describe two methods of following the rate of an acid-carbonate reaction and explain the graph obtained.

SLO: C-10-B-30, B-31

1. React a measured amount of calcium carbonate with acid and record time from mixing.
2. Method 1: collect CO_2 in a gas syringe and record volume at regular intervals.
3. Method 2: place an open, suitably covered flask on a balance and record decreasing mass as CO_2 escapes; do not seal an escaping-gas setup.
4. Average rate = volume increase/time, or mass decrease/time, with the corresponding units.
5. A gas-volume curve is initially steep, then flattens; the slope gives the rate at each stage.
6. A final horizontal section shows no further measurable gas formation. Compare runs using the same measurement method and comparable apparatus; see the rate graph.

L022 [6 marks] Gas volumes at 0, 20, 40, 60 and 80 s are 0, 24, 40, 48 and 48 cm^3. Calculate interval rates and interpret the pattern.

SLO: C-10-B-31

1. 0-20 s: rate = (24 - 0)/20 = 1.20 cm^3 s^{-1}.
2. 20-40 s: rate = (40 - 24)/20 = 0.80 cm^3 s^{-1}.
3. 40-60 s: rate = (48 - 40)/20 = 0.40 cm^3 s^{-1}.
4. 60-80 s: rate = (48 - 48)/20 = 0 cm^3 s^{-1}.
5. The rate decreases and no further gas is formed after 60 s; the final measured gas volume is 48 cm^3.
6. Average rate over the first 60 s = 48/60 = 0.80 cm^3 s^{-1}. Do not divide by 80 s when asked for the first 60 s.

L023 [6 marks] Curve A reaches 60 cm^3 gas in 40 s; curve B reaches 60 cm^3 in 75 s. Curve C reaches 30 cm^3 in 40 s. Compare average rates and final amounts.

SLO: C-10-B-31

1. Average rate A = 60/40 = 1.50 cm^3 s^{-1}.
2. Average rate B = 60/75 = 0.80 cm^3 s^{-1}.
3. Average rate C = 30/40 = 0.75 cm^3 s^{-1}.
4. A has the highest average rate over its stated interval, followed by B, then C.
5. A and B give equal amounts of gas at the same T and P; C gives half as much.
6. The data alone do not prove which experimental factor caused the differences. Initial rates require initial slopes, not just endpoint averages.

L024 [6 marks] Explain how rate measurements can help a food supplier choose harvest, storage and delivery schedules.

SLO: C-10-B-30, B-33

1. Food undergoes chemical changes during ripening and deterioration.
2. Measure a suitable quality change, such as gas production or another validated ripening indicator, over time.
3. Use the data to estimate how long produce takes to reach saleable ripeness and when quality becomes unacceptable.
4. Choose a harvesting stage that allows for the expected transport and sale time.
5. Use suitable cool storage and a reliable delivery schedule to slow deterioration and maintain quality.
6. Repeat measurements for the particular food and conditions, reducing spoilage and waste. Detailed collision theory, catalyst theory and factor explanations are outside the selected summative SLOs.

05 Salts

Included theory SLOs: C-10-B-34 to B-39

Short questions | 3 marks each

S065 [3 marks] What is a salt? Explain its ionic nature with an example.

SLO: C-10-B-34

1. A salt is an ionic compound containing positive and negative ions, commonly formed when the hydrogen ions of an acid are replaced by metal or ammonium ions.
2. The ions are held together by electrostatic attraction, and total positive and negative charges balance.
3. For example, NaCl contains Na^+ and Cl^-; in acid-base preparation Na^+ comes from NaOH and Cl^- from HCl.

S066 [3 marks] Why are salts normally crystalline solids with high melting points?

SLO: C-10-B-35, B-36

1. Their positive and negative ions form a regular giant ionic lattice.
2. Strong electrostatic attractions act between oppositely charged ions throughout the lattice.
3. A large amount of energy is needed to overcome these attractions, so salts generally remain solid at STP and have high melting points.

S067 [3 marks] Why does solid NaCl not conduct electricity while molten or aqueous NaCl does?

SLO: C-10-B-37

1. In solid NaCl, ions are fixed in the lattice and cannot carry charge through the solid.
2. In molten NaCl, ions are free to move to opposite electrodes.
3. In aqueous NaCl, dissolved ions are mobile and carry current; electrons are not the charge carriers through the solution.

S068 [3 marks] State solubility rules for sodium, potassium, ammonium and nitrate salts, with examples.

SLO: C-10-B-38

1. All common sodium and potassium salts are soluble, for example NaCl and K_2CO_3.
2. All common ammonium salts are soluble, for example NH_4Cl.
3. All common nitrates are soluble, for example $AgNO_3$ and $Pb(NO_3)_2$.

S069 [3 marks] State the solubility rules for chlorides and carbonates.

SLO: C-10-B-38

1. Most chlorides are soluble; silver chloride and lead(II) chloride are the main listed exceptions.
2. Most carbonates are insoluble.
3. Sodium, potassium and ammonium carbonates are soluble exceptions; for example Na_2CO_3 is soluble while $CaCO_3$ is insoluble.

S070 [3 marks] State the solubility rule for hydroxides and classify NaOH, $Cu(OH)_2$ and $Ca(OH)_2$.

SLO: C-10-B-38

1. Most metal hydroxides are insoluble; NaOH and KOH are soluble, and aqueous ammonia is often termed ammonium hydroxide in school rules.
2. NaOH is soluble, whereas $Cu(OH)_2$ is insoluble.
3. $Ca(OH)_2$ is only sparingly soluble; its clear dilute solution is limewater.

S071 [3 marks] Predict whether AgCl, K_2CO_3 and $Pb(NO_3)_2$ are soluble in water, giving reasons.

SLO: C-10-B-38

1. AgCl is insoluble: silver chloride is an exception to the chloride-solubility rule.
2. K_2CO_3 is soluble: all common potassium salts are soluble.
3. $Pb(NO_3)_2$ is soluble: all common nitrate salts are soluble.

S072 [3 marks] Why is excess insoluble base used when preparing a soluble salt from an acid?

SLO: C-10-B-39

1. Excess base ensures all acid is neutralised.
2. The unreacted insoluble base is removed by filtration.
3. The filtrate contains the salt solution, which can be concentrated and cooled to obtain crystals.

S073 [3 marks] Give an equation and three key steps for making copper(II) sulfate from copper(II) oxide.

SLO: C-10-B-39

1. Warm dilute sulfuric acid and add CuO until some black solid remains: $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$.
2. Filter off excess CuO to obtain blue copper(II) sulfate solution.
3. Gently evaporate some water, cool to crystallise, then filter and dry the crystals.

S074 [3 marks] Why is titration used to prepare NaCl from HCl and NaOH?

SLO: C-10-B-39

1. Both reactants are soluble, so excess alkali cannot be removed by filtration.
2. Titration with an indicator first identifies the exact neutralising volumes.
3. Repeat with those volumes without indicator, then concentrate and crystallise to obtain uncontaminated salt.

S075 [3 marks] Why should a salt solution be crystallised rather than always heated to dryness?

SLO: C-10-B-39

1. Gentle evaporation concentrates the solution until crystals can form on cooling.
2. Cooling then gives crystals, which are separated by filtration and dried.
3. Strong heating to dryness can decompose some salts or remove water of crystallisation, giving an unsuitable product.

S076 [3 marks] Give equations for preparing zinc sulfate with a metal and magnesium sulfate with a carbonate.

SLO: C-10-B-39

1. Metal route: $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$.
2. Carbonate route: $MgCO_3 + H_2SO_4 \rightarrow MgSO_4 + CO_2 + H_2O$.
3. Use the suitable solid in excess, filter away unreacted solid, and crystallise the soluble salt from the filtrate.

Long questions | 6 marks each

L025 [6 marks] Explain the structure, physical state, melting point and electrical conductivity of salts.

SLO: C-10-B-34 to B-37

1. Salts consist of positive and negative ions in charge-balanced proportions.
2. Oppositely charged ions are held by strong electrostatic attractions.
3. A regular three-dimensional arrangement forms a giant ionic lattice.
4. These strong attractions make salts generally crystalline solids at STP with high melting points.
5. Solid salts do not conduct because their ions cannot move through the lattice.
6. Molten salts and aqueous solutions of soluble salts conduct because mobile ions move toward opposite electrodes.

L026 [6 marks] State the prescribed solubility rules and apply them to $AgNO_3$, AgCl, $CaCO_3$ and KOH.

SLO: C-10-B-38

1. All common sodium, potassium and ammonium salts are soluble.
2. All common nitrates are soluble; therefore $AgNO_3$ is soluble.
3. Most chlorides are soluble except silver and lead(II) chlorides; AgCl is insoluble.
4. Most carbonates are insoluble except sodium, potassium and ammonium carbonates; $CaCO_3$ is insoluble.
5. Most hydroxides are insoluble; sodium and potassium hydroxides are soluble, and aqueous ammonia is treated as soluble in the school rule.
6. $Ca(OH)_2$ is sparingly soluble; KOH is soluble. State the rule as well as each classification.

L027 [6 marks] Describe preparation, separation and purification of copper(II) sulfate crystals using copper(II) oxide.

SLO: C-10-B-39

1. Warm dilute H_2SO_4 gently in a suitable container.
2. Add CuO in small portions with stirring until excess black solid remains; this ensures all acid has reacted.
3. Equation: $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$.
4. Filter to remove excess CuO; collect the blue salt solution as filtrate.
5. Evaporate some water gently to concentrate, then allow the solution to cool so hydrated crystals form.
6. Filter the crystals, rinse with a little cold distilled water and dry between filter papers. Do not strongly heat the crystals.

L028 [6 marks] Describe preparation of pure sodium chloride crystals from hydrochloric acid and sodium hydroxide.

SLO: C-10-B-39

1. Use a pipette to measure a known volume of NaOH into a flask and add a suitable indicator.
2. Add HCl from a burette while swirling until the endpoint is reached; record the volume.
3. Equation: $HCl + NaOH \rightarrow NaCl + H_2O$.
4. Repeat using the same measured volumes but no indicator, so the product is not contaminated by indicator.
5. Gently evaporate water from the neutral salt solution, then allow it to cool and crystallise.
6. Separate the crystals and dry them. Filtration of the reacting solution cannot remove excess soluble acid or alkali, so accurate neutralisation is needed.

L029 [6 marks] Compare preparation of a soluble salt using excess metal, insoluble base and insoluble carbonate.

SLO: C-10-B-39

1. Metal route: use a suitable metal such as Zn with dilute H_2SO_4; $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$.
2. Insoluble-base route: $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$.
3. Insoluble-carbonate route: $MgCO_3 + H_2SO_4 \rightarrow MgSO_4 + CO_2 + H_2O$.
4. In each method add the solid until some remains; for gas-producing routes the reaction stops when the acid is consumed.
5. Filter off unreacted solid and collect the salt solution.
6. Concentrate gently, cool, filter crystals and dry. Choose a suitable reactant: copper metal does not prepare $CuSO_4$ by reaction with dilute H_2SO_4.

06 Basics of Organic Chemistry

Included theory SLOs: C-10-E-01 to E-03

Short questions | 3 marks each

S077 [3 marks] Name the first four unbranched alkanes and write their structural formulae.

SLO: C-10-E-01, E-02

1. Methane: CH_4; ethane: CH_3-CH_3.
2. Propane: CH_3-CH_2-CH_3; butane: CH_3-CH_2-CH_2-CH_3.
3. These are alkanes, ending in -ane, with carbon-carbon single bonds; the open-chain general formula is CnH2n+2.

S078 [3 marks] Name and write structural formulae of ethene, propene and the two unbranched butenes.

SLO: C-10-E-01

1. Ethene: CH_2=CH_2; propene: CH_2=CH-CH_3.
2. But-1-ene: CH_2=CH-CH_2-CH_3.
3. But-2-ene: CH_3-CH=CH-CH_3. The locant gives the double bond its lowest possible position.

S079 [3 marks] Write the structural formulae of methanol, ethanol and the two propanols.

SLO: C-10-E-01

1. Methanol: CH_3-OH; ethanol: CH_3-CH_2-OH.
2. Propan-1-ol: CH_3-CH_2-CH_2-OH.
3. Propan-2-ol: CH_3-CH(OH)-CH_3; the -OH is attached to carbon 2.

S080 [3 marks] Write formulae of butan-1-ol and butan-2-ol, and state the difference.

SLO: C-10-E-01

1. Butan-1-ol: CH_3-CH_2-CH_2-CH_2-OH.
2. Butan-2-ol: CH_3-CH(OH)-CH_2-CH_3.
3. Both have molecular formula $C_4H_{10}O$, but the -OH group is on different carbon atoms, so their structures differ.

S081 [3 marks] Name and write structural formulae of the first four unbranched carboxylic acids.

SLO: C-10-E-01

1. Methanoic acid: H-C(=O)-OH; ethanoic acid: CH_3-C(=O)-OH.
2. Propanoic acid: CH_3-CH_2-C(=O)-OH; butanoic acid: CH_3-CH_2-CH_2-C(=O)-OH.
3. Each contains the carboxyl group -C(=O)-OH; include its carbon when counting the acid chain.

S082 [3 marks] Identify compound families from the endings -ane, -ene, -yne, -ol and -oic acid.

SLO: C-10-E-02

1. -ane identifies an alkane; -ene identifies an alkene.
2. -yne identifies an alkyne; -ol identifies an alcohol.
3. -oic acid identifies a carboxylic acid. These endings identify the family in the simple compounds covered here.

S083 [3 marks] Classify CH_3CH_3, $CH_2{=}CH_2$ and HC≡CH, giving the bonding reason.

SLO: C-10-E-02

1. CH_3CH_3 is ethane, an alkane, because its carbon-carbon bond is single.
2. $CH_2{=}CH_2$ is ethene, an alkene, because it has a carbon-carbon double bond.
3. HC≡CH is ethyne, an alkyne, because it has a carbon-carbon triple bond.

S084 [3 marks] Identify the families of CH_3CH_2OH, CH_3COOH and CH_3COOCH_3.

SLO: C-10-E-02

1. CH_3CH_2OH is an alcohol: ethanol, containing an -OH group attached to a saturated carbon.
2. CH_3COOH is a carboxylic acid: ethanoic acid, containing -COOH.
3. CH_3COOCH_3 is an ester: methyl ethanoate, containing -C(=O)-O- between carbon-containing groups.

S085 [3 marks] Distinguish molecular, condensed structural and displayed formulae using ethanol.

SLO: C-10-E-01, E-02

1. The molecular formula C_2H_6O gives only the number of atoms of each element.
2. The condensed structural formula CH_3-CH_2-OH shows the order in which groups are connected.
3. A displayed formula shows every atom and every covalent bond, including O-H and C-H bonds; see the structure atlas.

S086 [3 marks] Explain how an ester is named and name $HCOOCH_3$ and $CH_3COOCH_2CH_3$.

SLO: C-10-E-03

1. The alcohol-derived group is named first as an alkyl group, then the acid-derived part ends in -oate.
2. $HCOOCH_3$ is methyl methanoate: methyl from methanol and methanoate from methanoic acid.
3. $CH_3COOCH_2CH_3$ is ethyl ethanoate: ethyl from ethanol and ethanoate from ethanoic acid.

S087 [3 marks] Write structural formulae of methyl ethanoate, ethyl methanoate and propyl propanoate.

SLO: C-10-E-03

1. Methyl ethanoate: CH_3-C(=O)-O-CH_3.
2. Ethyl methanoate: H-C(=O)-O-CH_2-CH_3.
3. Propyl propanoate: CH_3-CH_2-C(=O)-O-CH_2-CH_2-CH_3; propyl here means propan-1-yl.

S088 [3 marks] Identify the parent alcohol and acid of butyl ethanoate and ethyl butanoate.

SLO: C-10-E-03

1. Butyl ethanoate has alcohol-derived butyl and acid-derived ethanoate groups.
2. Its parents are butan-1-ol and ethanoic acid: CH_3-C(=O)-O-CH_2-CH_2-CH_2-CH_3.
3. Ethyl butanoate has parents ethanol and butanoic acid: CH_3-CH_2-CH_2-C(=O)-O-CH_2-CH_3.

S089 [3 marks] Can the molecular formula C_4H_8 alone prove that a compound is but-1-ene? Explain.

SLO: C-10-E-01, E-02

1. No; a molecular formula gives atom numbers but not their connections.
2. Both but-1-ene and but-2-ene have formula C_4H_8.
3. A structural or displayed formula is needed to locate the double bond; molecular formulae can fit more than one structure or family.

S090 [3 marks] Name $CH_3CH{=}CHCH_3$, $CH_3CH(OH)CH_3$ and CH_3CH_2COOH.

SLO: C-10-E-01

1. $CH_3CH{=}CHCH_3$ is but-2-ene: four carbons, double bond beginning at carbon 2.
2. $CH_3CH(OH)CH_3$ is propan-2-ol: three carbons, -OH on carbon 2.
3. CH_3CH_2COOH is propanoic acid: three carbons including the carboxyl carbon.

Long questions | 6 marks each

L030 [6 marks] Draw displayed formulae and name methane, ethane, propane, butane, ethene and propene.

SLO: C-10-E-01

1. Methane is CH_4; display carbon joined by four single bonds to H.
2. Ethane is CH_3-CH_3; show a C-C bond and three C-H bonds on each carbon.
3. Propane is CH_3-CH_2-CH_3; the middle carbon has two hydrogen atoms.
4. Butane is CH_3-CH_2-CH_2-CH_3, an unbranched four-carbon chain.
5. Ethene is $CH_2{=}CH_2$; each carbon has two C-H bonds and one C=C bond.
6. Propene is $CH_2{=}CH$-CH_3. The fully displayed versions are provided in the structure atlas; each carbon has a total bond order of four.

L031 [6 marks] Draw and distinguish but-1-ene, but-2-ene, propan-1-ol, propan-2-ol, butan-1-ol and butan-2-ol.

SLO: C-10-E-01

1. But-1-ene: $CH_2{=}CH$-CH_2-CH_3, with a double bond starting at carbon 1.
2. But-2-ene: CH_3-$CH{=}CH$-CH_3, with a double bond starting at carbon 2.
3. Propan-1-ol: CH_3-CH_2-CH_2-OH.
4. Propan-2-ol: CH_3-CH(OH)-CH_3.
5. Butan-1-ol: CH_3-CH_2-CH_2-CH_2-OH; butan-2-ol: CH_3-CH(OH)-CH_2-CH_3.
6. Use the lowest locant for the named group or double bond. Fully displayed formulae of all six appear in the structure atlas.

L032 [6 marks] Identify six organic families by their names and characteristic structures, giving one example each.

SLO: C-10-E-01, E-02

1. Alkane: suffix -ane, only C-C single bonds in its carbon skeleton; ethane, CH_3-CH_3.
2. Alkene: suffix -ene, C=C bond; ethene, CH_2=CH_2.
3. Alkyne: suffix -yne, C≡C bond; ethyne, HC≡CH.
4. Alcohol: suffix -ol, hydroxyl group on a saturated carbon; ethanol, CH_3CH_2OH.
5. Carboxylic acid: suffix -oic acid, -C(=O)-OH group; ethanoic acid, CH_3COOH.
6. Ester: alkyl alkanoate name and -C(=O)-O- linkage; methyl ethanoate, CH_3COOCH_3.

L033 [6 marks] Name and draw displayed formulae of six esters: methyl methanoate, ethyl methanoate, methyl ethanoate, ethyl ethanoate, propyl propanoate and butyl butanoate.

SLO: C-10-E-03

1. Methyl methanoate: H-C(=O)-O-CH_3.
2. Ethyl methanoate: H-C(=O)-O-CH_2-CH_3.
3. Methyl ethanoate: CH_3-C(=O)-O-CH_3.
4. Ethyl ethanoate: CH_3-C(=O)-O-CH_2-CH_3.
5. Propyl propanoate: CH_3-CH_2-C(=O)-O-CH_2-CH_2-CH_3.
6. Butyl butanoate: CH_3-CH_2-CH_2-C(=O)-O-CH_2-CH_2-CH_2-CH_3. The ester atlas shows all atoms and bonds, including the carbonyl double bond.

L034 [6 marks] For the esters CH_3COOCH_3, $HCOOCH_2CH_3$ and $CH_3CH_2COOCH_2CH_3$, give names and identify the alcohol and acid represented by each.

SLO: C-10-E-03

1. CH_3COOCH_3 is methyl ethanoate.
2. Its alcohol-derived group is methyl from methanol, and its acid-derived group is ethanoate from ethanoic acid.
3. $HCOOCH_2CH_3$ is ethyl methanoate.
4. Its parent alcohol is ethanol and its parent acid is methanoic acid.
5. $CH_3CH_2COOCH_2CH_3$ is ethyl propanoate.
6. Its parent alcohol is ethanol and its parent acid is propanoic acid. The group on the single-bonded ester oxygen identifies the alcohol portion.

07 Hydrocarbons

Included theory SLOs: C-10-E-04 to E-14, including both entries printed E-12

Short questions | 3 marks each

S091 [3 marks] Why are alkenes called unsaturated hydrocarbons?

SLO: C-10-E-04

1. They contain carbon and hydrogen only, so they are hydrocarbons.
2. They contain at least one carbon-carbon double bond.
3. The double bond can undergo addition, so the molecule can take up more atoms; hence it is unsaturated.

S092 [3 marks] Compare alkenes and alkynes using bonds and examples.

SLO: C-10-E-04, E-10

1. Alkenes contain C=C; ethene is CH_2=CH_2.
2. Alkynes contain C≡C; ethyne is HC≡CH.
3. Both are unsaturated hydrocarbons; open-chain compounds with one such bond have formulae CnH2n and CnH2n-2 respectively.

S093 [3 marks] What is catalytic cracking? Give an equation and conditions.

SLO: C-10-E-05

1. Cracking breaks large alkane molecules into smaller useful molecules, including alkenes.
2. Vaporise the hydrocarbon and pass it over a hot silica/alumina or zeolite catalyst at a high temperature.
3. Example: $C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4$; the equation is balanced and produces an alkene.

S094 [3 marks] Why are large alkane molecules cracked in industry?

SLO: C-10-E-06

1. Cracking increases the supply of shorter-chain fuels such as petrol-range hydrocarbons.
2. It produces alkenes used as chemical feedstocks for plastics and other chemicals.
3. It converts less-demanded heavy fractions into more useful and valuable products.

S095 [3 marks] Describe the bromine-water test for unsaturation.

SLO: C-10-E-07

1. Add bromine water to the hydrocarbon and shake, away from strong UV light.
2. An alkene decolourises the orange-brown bromine water as bromine adds across C=C.
3. A saturated alkane does not rapidly decolourise it under these conditions; compare before and after observations.

S096 [3 marks] Describe the cold dilute alkaline $KMnO_4$ test for an alkene.

SLO: C-10-E-07

1. Shake the sample with cold dilute alkaline potassium manganate(VII), $KMnO_4$.
2. The purple colour disappears and brown MnO_2 can form as the alkene is oxidised to a diol.
3. An alkane does not show this rapid change under the same conditions. State the reagent conditions because the observation depends on the medium.

S097 [3 marks] Write the addition reaction of ethene with bromine and name the product.

SLO: C-10-E-08

1. $CH_2=CH_2 + Br_2 \rightarrow CH_2Br\text{-}CH_2Br$.
2. The product is 1,2-dibromoethane, with one bromine atom added to each carbon.
3. The C=C becomes C-C; bromine is decolourised.

S098 [3 marks] Describe hydrogenation of ethene.

SLO: C-10-E-08

1. Ethene reacts with hydrogen in the presence of a nickel catalyst, usually with heating.
2. $CH_2=CH_2 + H_2 \rightarrow CH_3\text{-}CH_3$.
3. The product is ethane; addition across C=C changes an unsaturated molecule into a saturated molecule.

S099 [3 marks] Describe hydration of ethene, including equation and conditions.

SLO: C-10-E-08

1. $CH_2=CH_2 + H_2O(g) \rightleftharpoons CH_3CH_2OH$.
2. Steam adds across the double bond to form ethanol.
3. Use an acid catalyst such as phosphoric acid at about 300°C and about 6000 kPa (approximately 60 atm).

S100 [3 marks] Write structural equations for adding bromine and hydrogen to propene.

SLO: C-10-E-08

1. Bromination: $CH_3\text{-}CH=CH_2 + Br_2 \rightarrow CH_3\text{-}CHBr\text{-}CH_2Br$.
2. The bromination product is 1,2-dibromopropane.
3. Hydrogenation: $CH_3\text{-}CH=CH_2 + H_2 \rightarrow CH_3\text{-}CH_2\text{-}CH_3$, using nickel and heat; the product is propane.

S101 [3 marks] How is ethene prepared from ethanol by elimination?

SLO: C-10-E-09

1. Heat ethanol with concentrated sulfuric acid at about 170°C.
2. $CH_3CH_2OH \rightarrow CH_2=CH_2 + H_2O$.
3. Water is eliminated, so the reaction is dehydration; the acid acts as a dehydrating agent.

S102 [3 marks] How is ethene prepared from bromoethane by elimination?

SLO: C-10-E-09

1. Heat bromoethane with ethanolic potassium hydroxide, usually under reflux.
2. $CH_3CH_2Br + KOH \rightarrow CH_2=CH_2 + KBr + H_2O$.
3. Hydrogen and bromine are removed from adjacent carbon atoms; use alcoholic KOH to favour elimination rather than aqueous KOH.

S103 [3 marks] Identify ethyne and explain the meaning of its triple bond.

SLO: C-10-E-10

1. Ethyne has molecular formula C_2H_2 and displayed arrangement H-C≡C-H.
2. The two carbon atoms share three pairs of electrons in a triple covalent bond.
3. It is an unsaturated hydrocarbon because the multiple bond can undergo addition.

S104 [3 marks] State the uses of ethyne in welding and artificial fruit ripening.

SLO: C-10-E-11

1. Ethyne burns in oxygen to produce a very hot oxy-acetylene flame used for welding and cutting metals.
2. Its combustion is $2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O$.
3. Ethyne can induce fruit ripening and has historically been used for artificial ripening; this is a syllabus use, not a recommendation to use impure carbide-derived gas on food.

S105 [3 marks] Explain the principle of fractional distillation of petroleum.

SLO: C-10-E-12a

1. Petroleum is a mixture of hydrocarbons with different boiling ranges.
2. Heated petroleum vapours enter a fractionating column that is hot at the bottom and cooler at the top.
3. Fractions condense at different heights according to boiling range; this is physical separation, not cracking.

S106 [3 marks] State how four properties change from the bottom to the top of a petroleum fractionating column.

SLO: C-10-E-12b

1. Hydrocarbon chain length decreases and boiling points decrease.
2. Volatility increases: top fractions vaporise more readily.
3. Viscosity decreases: top fractions flow more easily.

S107 [3 marks] Give the uses of refinery gas, petrol and naphtha.

SLO: C-10-E-14

1. Refinery gas is used for heating and cooking.
2. Petrol/gasoline is used as fuel in cars.
3. Naphtha is used as a chemical feedstock to manufacture other substances.

S108 [3 marks] Give the uses of kerosene, diesel oil and fuel oil.

SLO: C-10-E-14

1. Kerosene/paraffin is used as jet fuel.
2. Diesel oil/gas oil is used in diesel engines.
3. Fuel oil is used in ships and in suitable heating systems.

S109 [3 marks] State uses of lubricating oil and bitumen, and explain their position in the column.

SLO: C-10-E-14

1. Lubricating-oil fractions provide lubricants, waxes and polishes.
2. Bitumen is used in road surfacing.
3. They are heavy, high-boiling fractions obtained low in the column; bitumen remains as residue at the bottom.

S110 [3 marks] Distinguish cracking from fractional distillation.

SLO: C-10-E-05, E-12a

1. Fractional distillation separates an existing mixture according to boiling ranges.
2. Cracking is a chemical change that breaks large hydrocarbon molecules into smaller molecules.
3. Distillation does not change molecular structures; cracking forms new substances and commonly produces alkenes.

Long questions | 6 marks each

L035 [6 marks] Describe catalytic cracking and explain its industrial importance.

SLO: C-10-E-05, E-06

1. Large hydrocarbons from heavy petroleum fractions are first vaporised.
2. The vapour passes over a hot suitable catalyst such as silica/alumina or a zeolite.
3. High temperature breaks some carbon-carbon bonds to form smaller molecules.
4. One possible balanced equation is $C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4$.
5. Shorter hydrocarbons help meet demand for useful fuels such as petrol.
6. Alkenes are valuable chemical feedstocks for making plastics and other products; cracking upgrades less-demanded heavy fractions.

L036 [6 marks] Explain two tests to distinguish ethene from ethane, including reagents, observations and interpretation.

SLO: C-10-E-04, E-07

1. Test 1: shake separate samples with bromine water away from strong UV light.
2. Ethene rapidly decolourises the orange-brown reagent; ethane shows no rapid change.
3. Ethene adds bromine across C=C: $CH_2=CH_2 + Br_2 \rightarrow CH_2Br\text{-}CH_2Br$.
4. Test 2: shake with cold dilute alkaline $KMnO_4$.
5. Ethene removes the purple colour, often producing brown MnO_2; ethane has no rapid reaction under these conditions.
6. Both tests indicate unsaturation in this comparison. Keep conditions the same and specify the $KMnO_4$ medium; many other reducing substances can also react with these reagents.

L037 [6 marks] Describe three addition reactions of ethene with structural equations, products and conditions.

SLO: C-10-E-08

1. With bromine: $CH_2=CH_2 + Br_2 \rightarrow CH_2Br\text{-}CH_2Br$.
2. The product is 1,2-dibromoethane and orange-brown bromine is decolourised.
3. With hydrogen: $CH_2=CH_2 + H_2 \rightarrow CH_3\text{-}CH_3$.
4. The product is ethane; use a nickel catalyst with heat.
5. With steam: $CH_2=CH_2 + H_2O(g) \rightleftharpoons CH_3CH_2OH$.
6. The product is ethanol; use phosphoric acid, about 300°C and about 6000 kPa. In each case the C=C double bond becomes a C-C single bond.

L038 [6 marks] Explain preparation of ethene by elimination from an alcohol and a halogenoalkane.

SLO: C-10-E-09

1. Alcohol route: use ethanol as the starting compound.
2. Heat with concentrated H_2SO_4 at about 170°C; $CH_3CH_2OH \rightarrow CH_2=CH_2 + H_2O$.
3. This is dehydration: water is eliminated from the alcohol.
4. Halogenoalkane route: heat bromoethane with ethanolic KOH under reflux.
5. $CH_3CH_2Br + KOH \rightarrow CH_2=CH_2 + KBr + H_2O$.
6. Hydrogen and bromine are removed from adjacent carbons and a C=C bond forms. Confirm the ethene using decolourisation of bromine water.

L039 [6 marks] Describe the structure of ethyne and explain its syllabus applications.

SLO: C-10-E-10, E-11

1. Ethyne is C_2H_2 with displayed formula H-C≡C-H.
2. It contains a carbon-carbon triple covalent bond.
3. It contains only carbon and hydrogen and is unsaturated.
4. It burns in oxygen: $2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O$.
5. The oxy-acetylene flame is very hot and is used in metal welding and cutting.
6. Ethyne can stimulate fruit ripening and has been used artificially for this purpose. This identifies the curriculum use; it does not establish that carbide-based ripening is safe.

L040 [6 marks] Explain fractional distillation of petroleum and the changes in fraction properties up the column.

SLO: C-10-E-12a, E-12b

1. Heat petroleum so that much of it vaporises and introduce the hot mixture into a fractionating column.
2. The column is hottest at the bottom and coolest at the top.
3. Vapours rise and condense in regions cool enough for their boiling ranges; high-boiling fractions collect lower down.
4. Low-boiling fractions rise farther and condense near the top; refinery gases leave at the top and bitumen remains at the bottom.
5. From bottom to top: chain length decreases, boiling point decreases, volatility increases and viscosity decreases.
6. The process physically separates mixtures into fractions with ranges of boiling points; it does not split large molecules. See the labelled column diagram.

L041 [6 marks] Name the eight prescribed petroleum fractions and state one use of each.

SLO: C-10-E-14

1. Refinery gas: heating and cooking; petrol/gasoline: car fuel.
2. Naphtha: chemical feedstock.
3. Kerosene/paraffin: jet fuel.
4. Diesel oil/gas oil: fuel for diesel engines.
5. Fuel oil: ships and suitable heating systems.
6. Lubricating oil: lubricants, waxes and polishes; bitumen: road surfacing. These eight fraction-use pairs are listed together in the reference table.

L042 [6 marks] Starting with ethanol, show how to obtain ethene and then convert separate samples of ethene into ethane, 1,2-dibromoethane and ethanol.

SLO: C-10-E-08, E-09

1. Dehydrate ethanol: $CH_3CH_2OH \rightarrow CH_2{=}CH_2 + H_2O$.
2. Use concentrated H_2SO_4 and heat at about 170°C for this elimination step.
3. Hydrogenate ethene: $CH_2{=}CH_2 + H_2 \rightarrow CH_3CH_3$, using nickel and heat.
4. Brominate ethene: $CH_2{=}CH_2 + Br_2 \rightarrow CH_2Br\text{-}CH_2Br$; bromine is decolourised.
5. Hydrate ethene: $CH_2{=}CH_2 + H_2O(g) \rightleftharpoons CH_3CH_2OH$, using phosphoric acid at about 300°C and 6000 kPa.
6. The first reaction is elimination; the other three are addition reactions at C=C. State conditions as well as the structural equations.

08 Hydroxy Compounds

Included theory SLOs: C-10-E-15 to E-19

Short questions | 3 marks each

S111 [3 marks] Give the equation and conditions for manufacturing ethanol by fermentation.

SLO: C-10-E-15

1. $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$.
2. Use aqueous glucose with yeast at about 25-35°C.
3. Exclude oxygen; enzymes in yeast catalyse the fermentation.

S112 [3 marks] Why are yeast, a suitable temperature and absence of oxygen needed in fermentation?

SLO: C-10-E-15

1. Yeast provides enzymes that convert glucose into ethanol and carbon dioxide.
2. About 25-35°C supports enzyme activity; excessive heat denatures the enzymes and can kill yeast.
3. Oxygen is excluded to favour fermentation rather than aerobic respiration.

S113 [3 marks] Describe manufacture of ethanol by hydration of ethene.

SLO: C-10-E-15

1. Ethene reacts with steam: $C_2H_4 + H_2O(g) \rightleftharpoons C_2H_5OH$.
2. Use a phosphoric acid catalyst at about 300°C and 6000 kPa, approximately 60 atm.
3. Cool to recover ethanol and recycle unreacted feed gases; the starting alkene is ethene, not ethane.

S114 [3 marks] Give three differences between fermentation and ethene hydration.

SLO: C-10-E-15

1. Fermentation uses renewable glucose; conventional hydration uses ethene commonly obtained from finite petroleum.
2. Fermentation occurs at 25-35°C and low pressure; hydration requires about 300°C and high pressure.
3. Fermentation is relatively slow and gives dilute ethanol needing purification; hydration can run rapidly and continuously.

S115 [3 marks] Write balanced complete-combustion equations for methanol and ethanol.

SLO: C-10-E-16

1. Methanol: $2CH_3OH + 3O_2 \rightarrow 2CO_2 + 4H_2O$.
2. Ethanol: $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$.
3. Both are exothermic reactions requiring sufficient oxygen; the released energy makes alcohols useful fuels.

S116 [3 marks] Compare complete and incomplete combustion of ethanol.

SLO: C-10-E-16

1. With sufficient oxygen, complete combustion produces CO_2 and H_2O: $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$.
2. Limited oxygen can produce toxic carbon monoxide: $C_2H_5OH + 2O_2 \rightarrow 2CO + 3H_2O$.
3. Severe oxygen shortage can also produce soot; incomplete combustion wastes useful fuel energy and gives harmful products.

S117 [3 marks] Give two advantages and one disadvantage of ethanol as a fuel.

SLO: C-10-E-17

1. Ethanol made from plant material can be renewable if feedstocks are replenished sustainably.
2. Its oxygen content can help cleaner combustion and reduce some pollutants in suitable fuel blends.
3. It has lower energy per unit volume than petrol, so a larger volume is needed for the same useful energy; production also uses land and energy.

S118 [3 marks] Explain three industrial uses of alcohols.

SLO: C-10-E-18

1. Pharmaceuticals: ethanol acts as a solvent in suitable medicines and in extraction processes.
2. Cosmetics: ethanol is a solvent or carrier in perfumes, sprays and some personal-care products.
3. Fuel production: ethanol can be blended with petrol, while some alcohols are industrial chemical feedstocks.

S119 [3 marks] Why is ethanol useful as a solvent and disinfectant?

SLO: C-10-E-19

1. Its molecule has both a polar -OH part and a non-polar carbon-containing part, allowing it to dissolve many useful substances.
2. This makes it useful in formulations such as perfumes and some pharmaceutical preparations.
3. Suitable aqueous ethanol formulations disinfect by disrupting membranes and denaturing proteins; around 70% ethanol is commonly used, and pure ethanol is not automatically more effective.

S120 [3 marks] State three limitations or precautions associated with alcohol use in daily life.

SLO: C-10-E-17, E-19

1. Alcohols such as ethanol are flammable and must be kept away from ignition sources.
2. Methanol is highly toxic; industrial alcohol and disinfectant formulations must not be swallowed.
3. Bioethanol production uses land, water and processing energy; it is not automatically free of environmental impacts.

Long questions | 6 marks each

L043 [6 marks] Describe manufacture of ethanol by fermentation, including conditions, equation and product recovery.

SLO: C-10-E-15

1. Prepare an aqueous glucose solution and add yeast.
2. Maintain about 25-35°C so yeast enzymes function effectively.
3. Exclude oxygen to favour anaerobic fermentation.
4. Equation: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$.
5. Fermentation is relatively slow and produces a dilute ethanol solution; increasing ethanol concentration eventually inhibits yeast.
6. Separate and concentrate the ethanol by distillation. The starting glucose can come from renewable plant material, but processing still needs energy.

L044 [6 marks] Describe ethanol manufacture by ethene hydration and compare it with fermentation.

SLO: C-10-E-15

1. Hydration equation: $C_2H_4 + H_2O(g) \rightleftharpoons C_2H_5OH$.
2. Use steam, phosphoric acid catalyst, about 300°C and about 6000 kPa (approximately 60 atm).
3. Hydration is fast and can be continuous; unreacted gases can be recycled.
4. Conventional ethene supply is commonly petroleum-based and finite, while glucose for fermentation can be renewable.
5. Fermentation uses yeast at 25-35°C without oxygen, so reactor temperature and pressure are lower.
6. Fermentation is slower and produces dilute ethanol requiring energy for separation; hydration requires hotter, pressurised equipment. Choose by feedstock, energy demand, purity and production scale.

L045 [6 marks] Explain combustion of alcohols and evaluate ethanol as a fuel compared with petrol.

SLO: C-10-E-16, E-17

1. Complete ethanol combustion: $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$; energy is released.
2. Complete methanol combustion: $2CH_3OH + 3O_2 \rightarrow 2CO_2 + 4H_2O$.
3. Plant-derived ethanol can be renewable and used in suitable petrol blends.
4. Its oxygen content can reduce some incomplete-combustion emissions in suitable engines, but combustion still releases CO_2.
5. Ethanol has lower energy per unit volume than petrol and can require compatible engine and fuel-system materials.
6. Crop production and distillation use land, water and energy, so lifecycle carbon savings depend on production methods; alcohol fuels are not automatically carbon-neutral.

L046 [6 marks] Explain the roles of alcohols in pharmaceuticals, cosmetics, fuels and daily life, including limitations.

SLO: C-10-E-18, E-19

1. Ethanol dissolves many compounds and is used as a solvent in suitable medicines and extraction.
2. It carries fragrance ingredients in perfumes and serves as a solvent in cosmetics and sprays.
3. Ethanol is used in motor-fuel blends and some heating or cooking fuels.
4. Suitable aqueous alcohol formulations disinfect surfaces by disrupting microbial membranes and proteins; they are not equally effective against every organism.
5. Alcohols are flammable, and industrial alcohol products must not be swallowed; methanol is particularly toxic.
6. Benefits therefore depend on the correct alcohol, concentration and purpose, with suitable storage and handling.

L047 [6 marks] A producer can obtain cheap glucose or ethene. Compare the ethanol routes and explain why neither fuel route is automatically carbon-neutral.

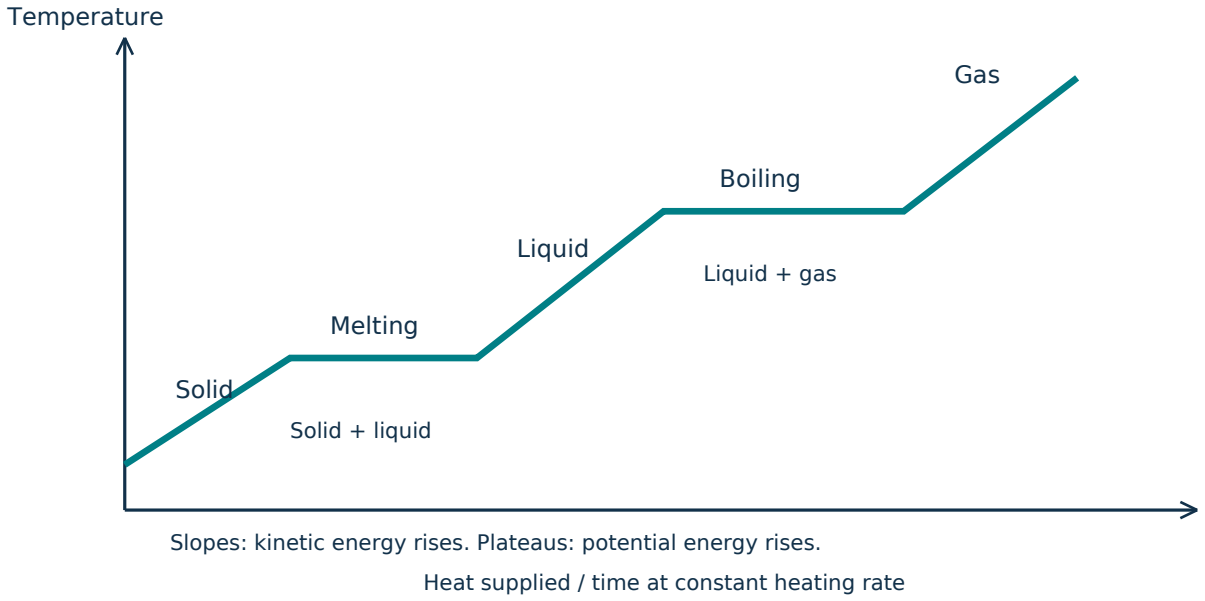
SLO: C-10-E-15 to E-17

1. Glucose route: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$, using yeast at 25-35°C without oxygen.
2. Ethene route: $C_2H_4 + H_2O(g) \rightleftharpoons C_2H_5OH$, using phosphoric acid at about 300°C and 6000 kPa.
3. Fermentation uses potentially renewable crops but is slow and gives dilute ethanol needing separation.
4. Hydration can be fast and continuous but conventional ethene comes from finite petroleum and requires high-pressure equipment.
5. Ethanol combustion releases CO_2: $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$.
6. Crop growth can absorb CO_2, but farming, fertiliser manufacture, transport and distillation also cause emissions; assess the whole production cycle before claiming carbon neutrality.

Diagram and graph reference

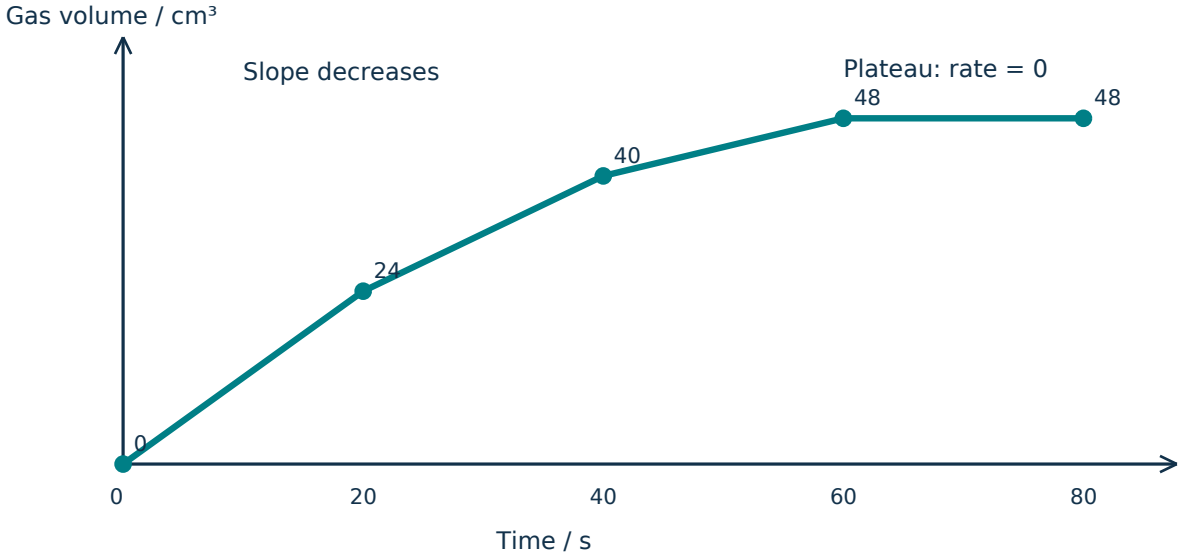
Use clear labels and relevant explanations in the exam. These diagrams support the corresponding questions; they are not additional SLOs.

1. Heating curve of a pure substance



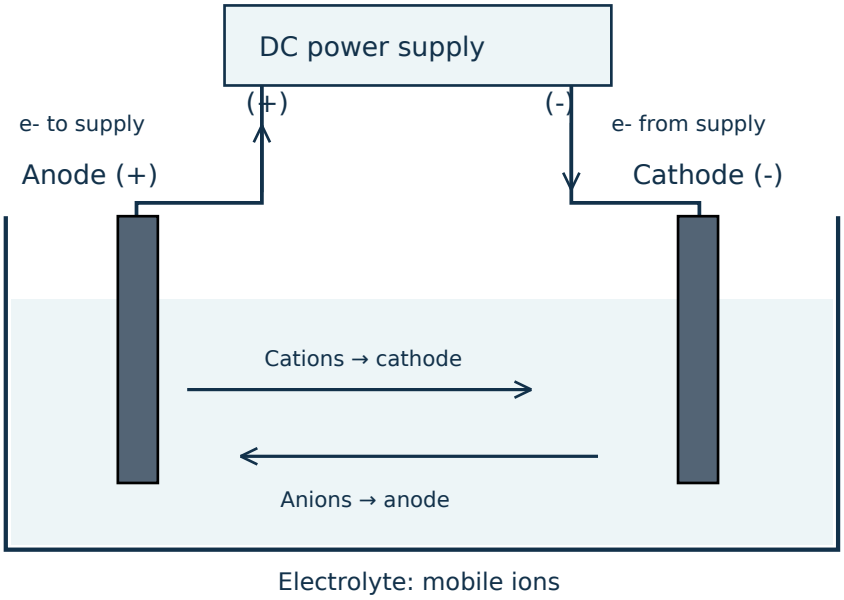
Temperature changes along sloping sections. At constant pressure, pure-substance phase changes produce plateaus; absorbed heat increases potential energy. A cooling curve runs through the reverse stages.

2. Gas-volume versus time graph



Data: 0, 24, 40, 48, 48 cm^3 at 0, 20, 40, 60, 80 s. The plotted line segments show interval averages; the slope of a tangent to a smooth experimental curve gives the instantaneous rate.

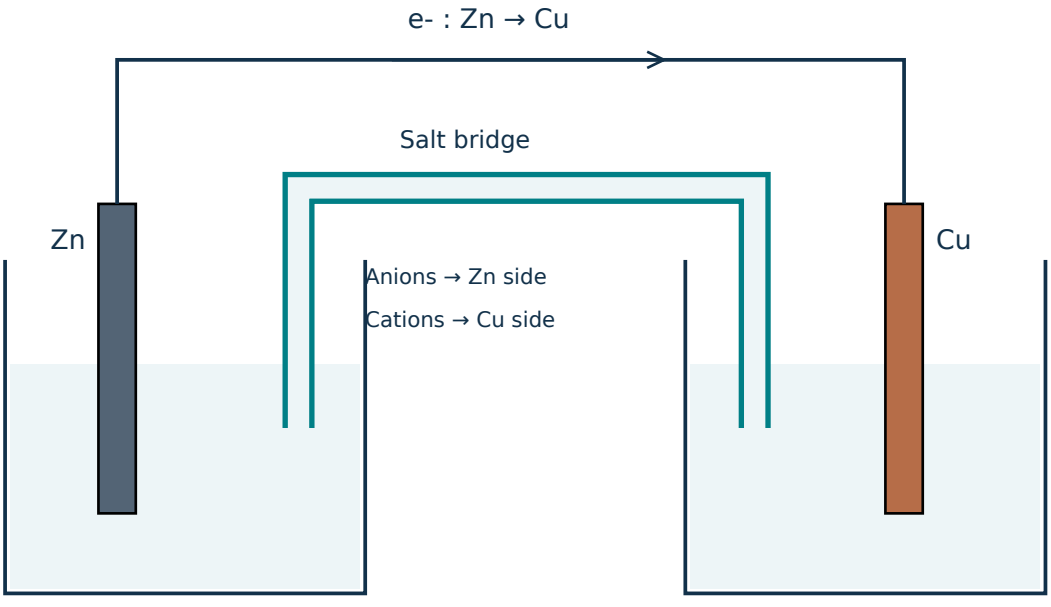
3. Electrolytic cell and electroplating arrangement



Electroplating: replace cathode by object; use plating metal as anode.

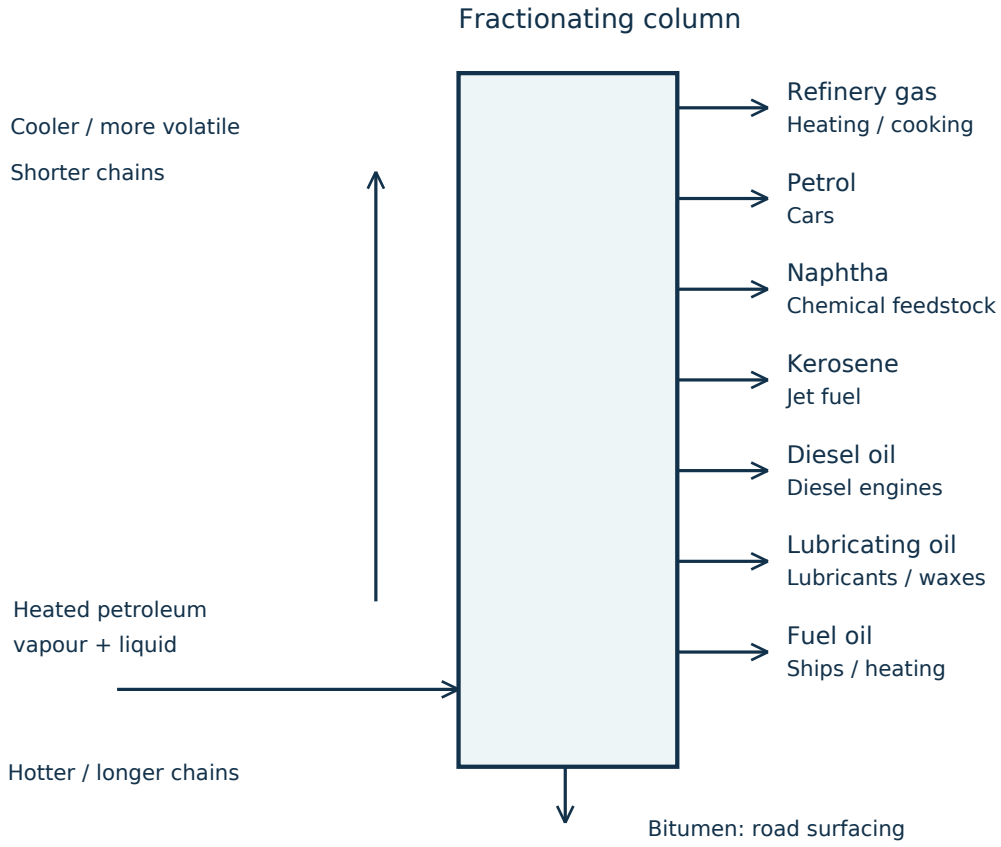
Electrolysis: anode positive, cathode negative. Oxidation releases electrons at the anode; reduction consumes electrons at the cathode. For copper plating, use $CuSO_4$ solution, a copper anode and the object as cathode.

4. Daniell (voltaic) cell



Cell notation: Zn(s) | Zn^{2+}(aq) || Cu^{2+}(aq) | Cu(s). The salt bridge carries ions, not electrons. Its arrows show ion movement needed to preserve charge balance.

5. Petroleum fractional distillation



Schematic only; fractions have overlapping boiling ranges and exact refinery cuts vary. The essential tested trend from bottom to top is shorter chains, lower boiling point, higher volatility and lower viscosity. Fuel and lubricating-oil ranges can overlap; learn each named use.

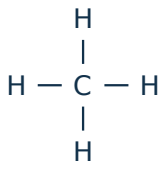
Fraction	Prescribed use
Refinery gas	Heating and cooking
Petrol / gasoline	Car fuel
Naphtha	Chemical feedstock
Kerosene / paraffin	Jet fuel
Diesel oil / gas oil	Diesel engines
Fuel oil	Ships and suitable heating systems
Lubricating oil	Lubricants, waxes and polishes
Bitumen	Road surfacing

Displayed-formula atlas

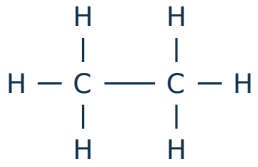
Each diagram explicitly shows all atoms and covalent bonds. Single, double and triple bonds must be drawn correctly. In condensed formulae, CH(OH) places -OH on the indicated carbon.

Alkanes | E-01

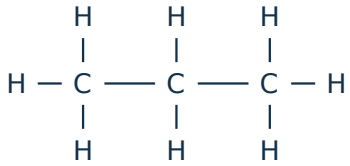
Methane



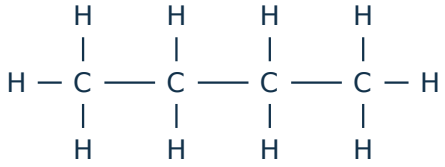
Ethane



Propane



Butane

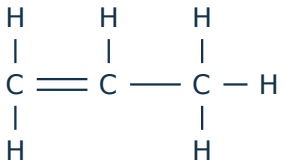


Alkenes | E-01

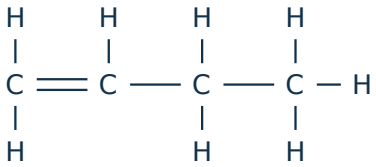
Ethene



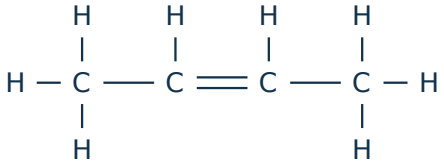
Propene



But-1-ene

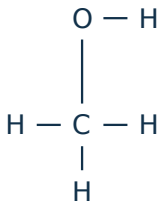


But-2-ene

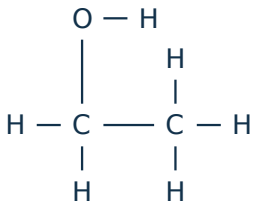


Alcohols | E-01

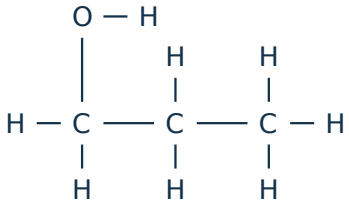
Methanol



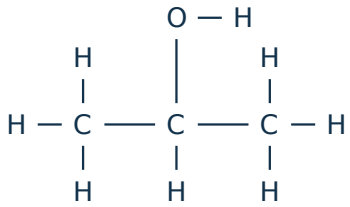
Ethanol



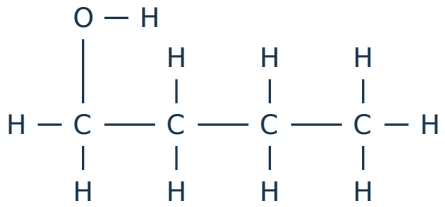
Propan-1-ol



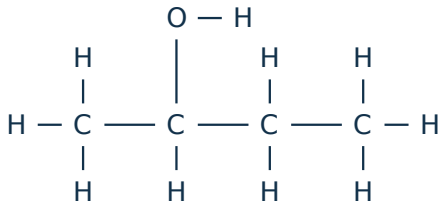
Propan-2-ol



Butan-1-ol

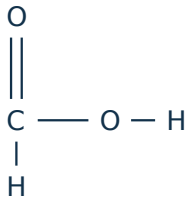


Butan-2-ol

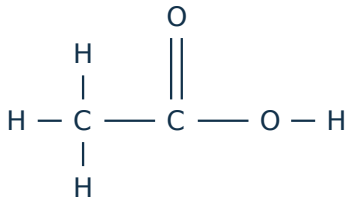


Carboxylic acids | E-01

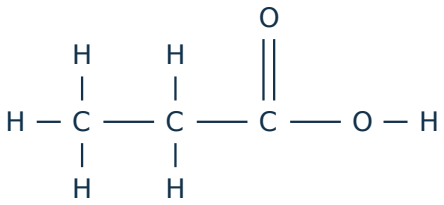
Methanoic acid



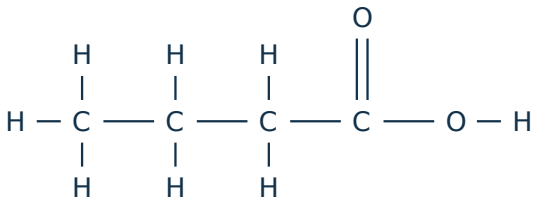
Ethanoic acid



Propanoic acid



Butanoic acid



Ethyne displayed formula: H-C≡C-H. Each carbon has one C-H bond and one C≡C bond.

Straight-chain naming extension | E-01 and E-02

E-01 gives no upper carbon limit for the first naming outcome. Use these parent stems and the same lowest-locant rules for longer unbranched chains; the explicitly listed positional examples are displayed above.

Carbon atoms	Alkane	1-alkene (where possible)	Primary alcohol / acid
1	methane	—	methanol / methanoic acid
2	ethane	ethene	ethanol / ethanoic acid
3	propane	propene	propan-1-ol / propanoic acid
4	butane	but-1-ene	butan-1-ol / butanoic acid
5	pentane	pent-1-ene	pentan-1-ol / pentanoic acid
6	hexane	hex-1-ene	hexan-1-ol / hexanoic acid
7	heptane	hept-1-ene	heptan-1-ol / heptanoic acid
8	octane	oct-1-ene	octan-1-ol / octanoic acid
9	nonane	non-1-ene	nonan-1-ol / nonanoic acid
10	decane	dec-1-ene	decan-1-ol / decanoic acid

For an internal double bond, state its locant, for example pent-2-ene: CH_3-CH=CH-CH_2-CH_3. For an internal -OH, state its locant, for example pentan-2-ol: CH_3-CH(OH)-CH_2-CH_2-CH_3. Naming and drawing any new example follows the same rules.

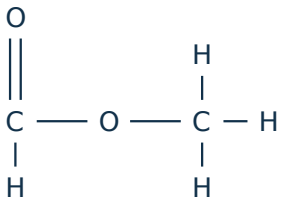
Four drawing checks

1. Choose the correct number of carbon atoms. 2. Place the multiple bond or functional group at the named position. 3. Complete each carbon to total bond order four and oxygen to two. 4. Show every hydrogen if a displayed formula is requested.

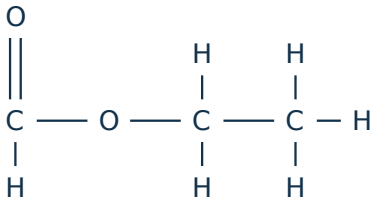
Ester atlas | E-03

The following 16 esters cover the four straight-chain primary alcohols and four unbranched carboxylic acids containing one to four carbons each. The carbon limit applies to each parent, not the total carbon number of the ester.

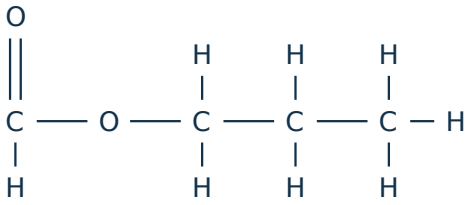
methyl methanoate



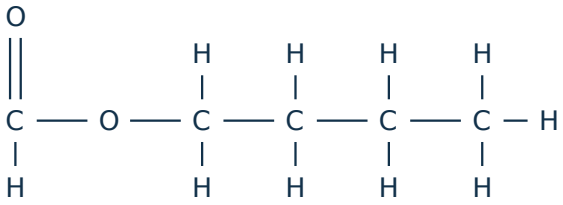
ethyl methanoate



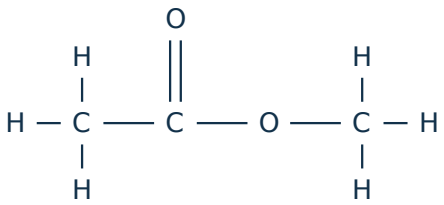
propyl methanoate



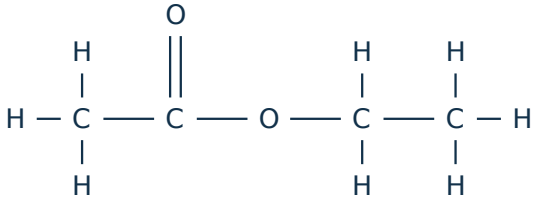
butyl methanoate



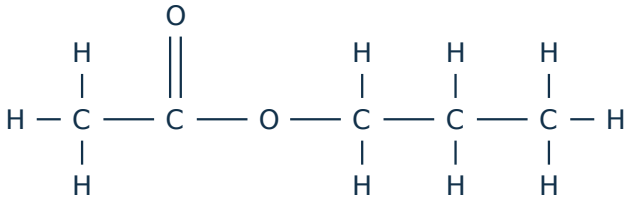
methyl ethanoate



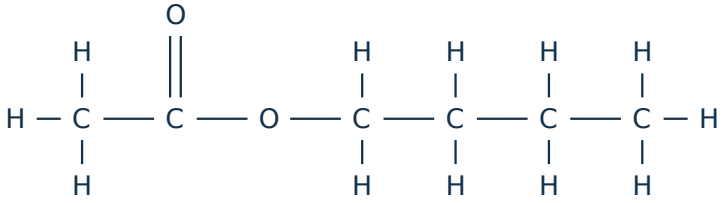
ethyl ethanoate



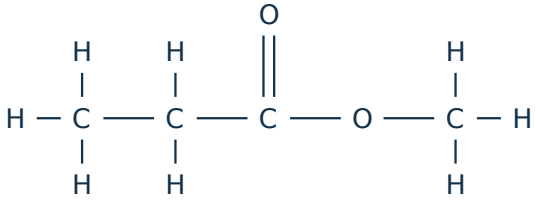
propyl ethanoate



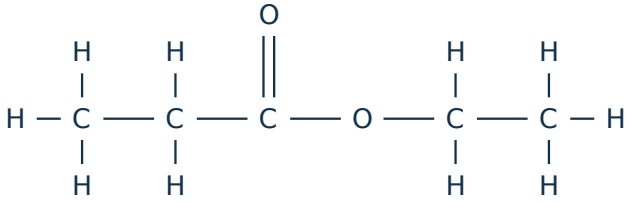
butyl ethanoate



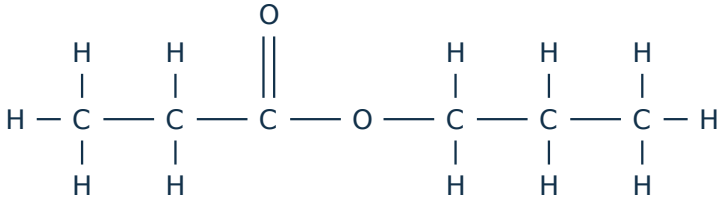
methyl propanoate



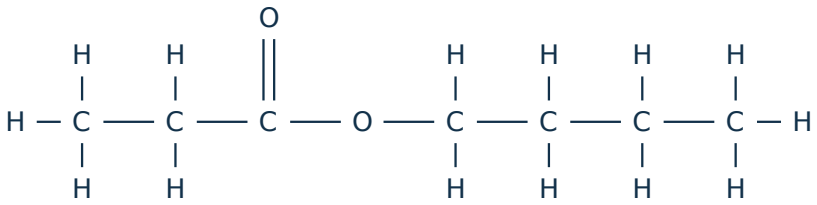
ethyl propanoate



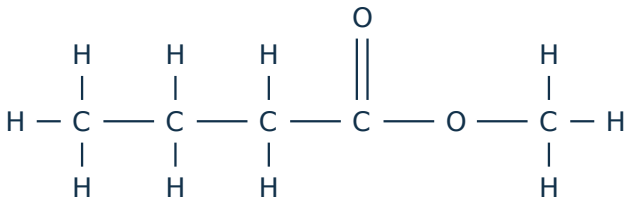
propyl propanoate



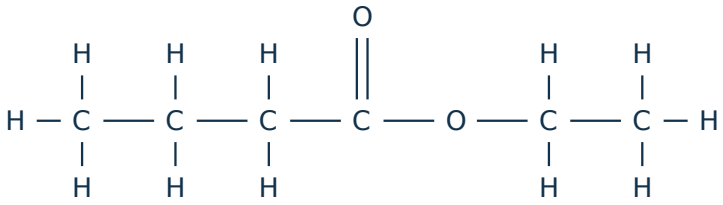
butyl propanoate



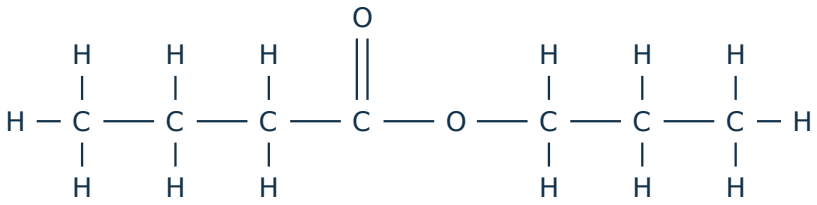
methyl butanoate



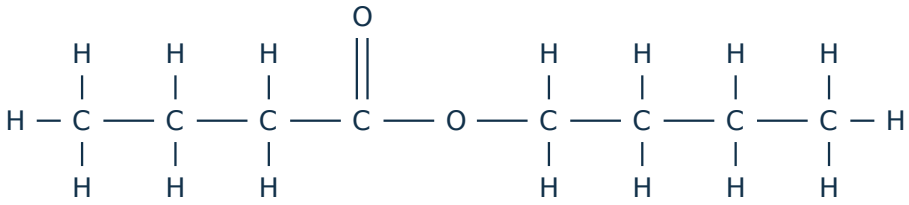
ethyl butanoate



propyl butanoate



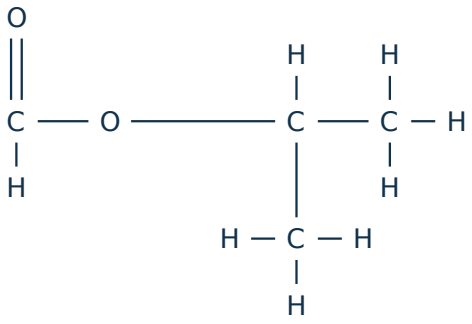
butyl butanoate



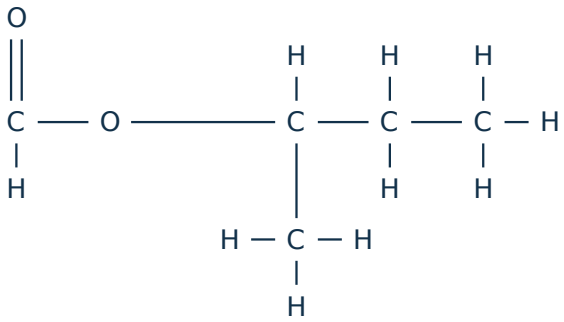
Secondary-alcohol ester extension | E-03

The source also explicitly names propan-2-ol and butan-2-ol in E-01. For completeness, these eight esters cover those unbranched secondary-alcohol parents with the same four acids. The alcohol skeleton is drawn with its first carbon below the second to show the O-carbon-2 connection clearly.

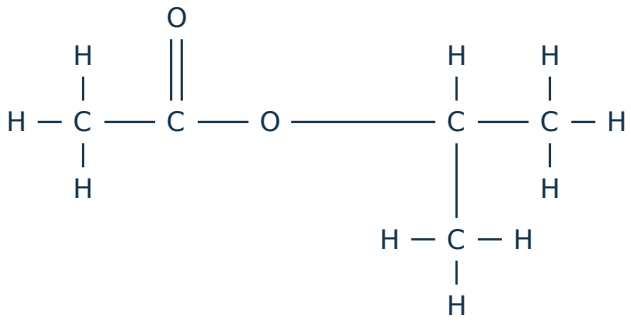
propan-2-yl methanoate



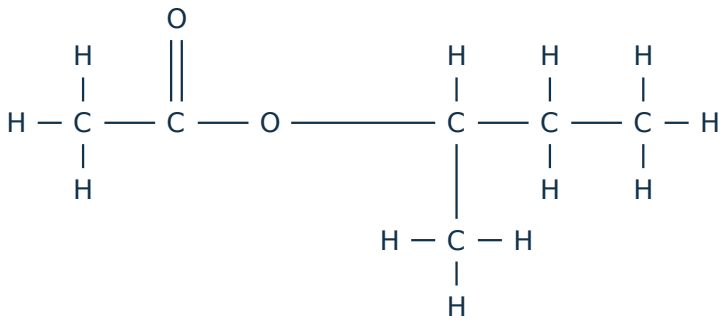
butan-2-yl methanoate



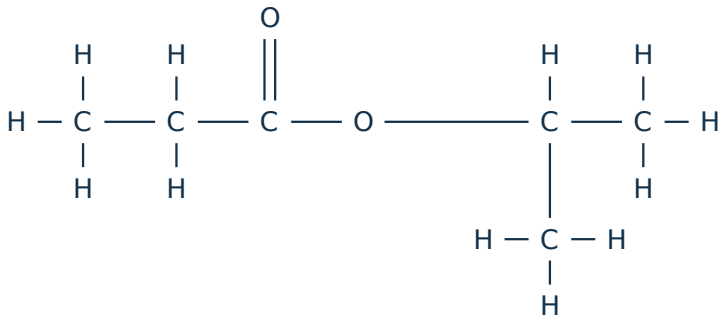
propan-2-yl ethanoate



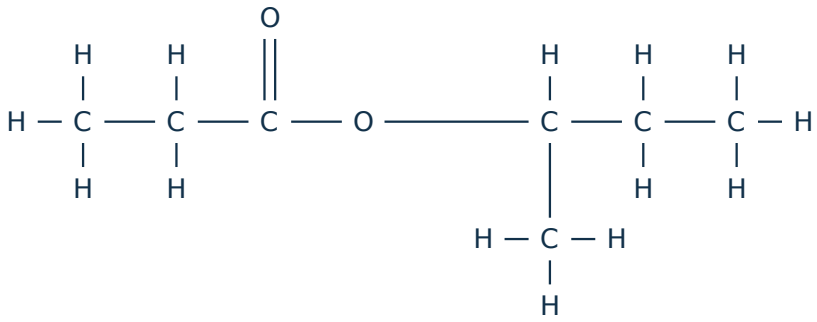
butan-2-yl ethanoate



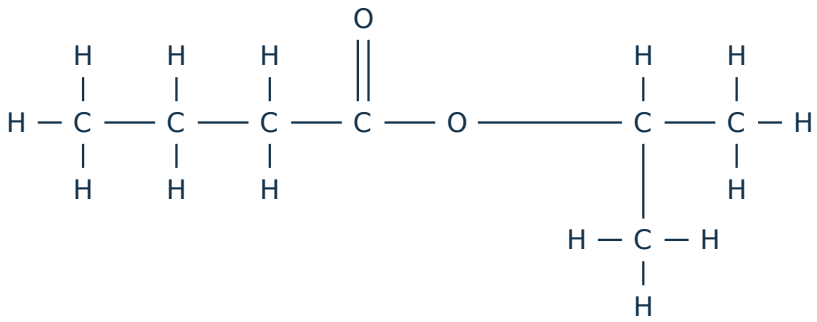
propan-2-yl propanoate



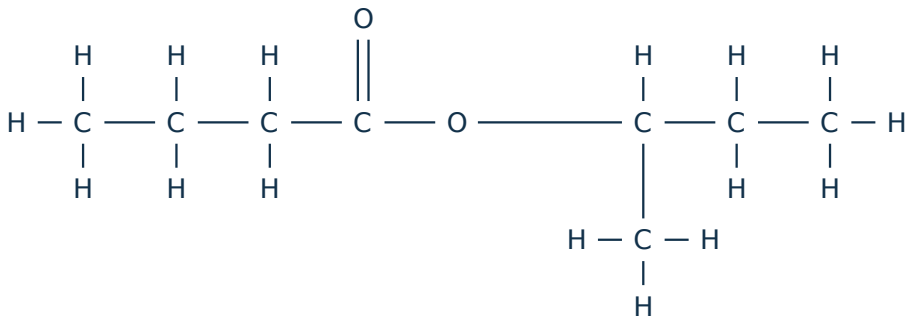
butan-2-yl propanoate



propan-2-yl butanoate



butan-2-yl butanoate



SLO coverage index

This index lists only the selected summative theory SLOs. Use the stable question IDs to locate answers. The diagram section and atlas add visual support.

C-10-B-01

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Final revision checklist

Matter

Explain energy changes at phase-change plateaus; distinguish evaporation and boiling; state the fixed conditions in each gas law; use kelvin; connect diffusion to particle motion.

Stoichiometry

Balance the equation first. Convert cm^3 to dm^3 before using $n = cV$. Use $m = nM$. Apply mole ratios. State units. For empirical formulae divide masses by Ar, then simplify the mole ratio.

Electrochemistry

Identify molten versus aqueous electrolyte and inert versus active electrodes. Use the correct electrode signs for the cell type. Balance atoms and charge in half-equations. Electrons carry current in wires; ions carry it in electrolyte.

Kinetics

Read graph axes and units. Rate is a change divided by time. Compare slopes for rates and plateaus for final amounts. Do not confuse greater final gas volume with a faster initial rate.

Salts

Use the stated solubility rules. Select a suitable salt-preparation method. Remove excess insoluble solid by filtration, then crystallise the filtrate. Repeat a titration preparation without indicator to avoid contamination.

Organic basics

Read the ending and locate the functional group. Count the acid carbonyl carbon. Draw the C=O bond in acids and esters. For esters, name the alcohol-derived alkyl group first.

Hydrocarbons

Distinguish physical separation from chemical cracking. Learn addition and elimination equations, conditions and product structures. State bromine/$KMnO_4$ observations with reagent conditions.

Hydroxy compounds

Learn both ethanol-manufacture equations and conditions. Use ethene, not ethane, for hydration. Balance combustion equations. Compare benefits with feedstock, energy demand and practical limitations.

Source and editorial notes

Primary scope source: user-supplied “FBISE_Chemistry_Class_10_Summative_and_Formative_Topics(1).pdf”, National Curriculum of Pakistan 2022-2023, Assessment Framework Chemistry Grade-X (SSC-II), printed pages 2-6 and 8-10. Assessment labels, not chapter titles alone, determine inclusion.

Technical cross-check for corrected hydration conditions: Pearson Edexcel International Advanced Level Chemistry, WCH_{12}/01, March 2021, question 19 (ethene + steam, phosphoric acid, 300°C and 60 atm). https://qualifications.pearson.com/content/dam/pdf/International%20Advanced%20Level/Chemistry/2018/Exam-materials/WCH_{12}_01_que_20210304.pdf

Technical cross-check for competing aqueous electrolysis reactions: University of Calgary Chemistry Textbook, “The Electrolysis of Aqueous Sodium Chloride”. https://chem-textbook.ucalgary.ca/version2/chapter-17-electrochemistry-introduction/electrolysis/the-electrolysis-of-aqueous-sodium-chloride/

The questions and model answers are independently prepared practice material. The framework does not supply a marking scheme or evidence of future question likelihood; no claim of an official answer key or guaranteed full marks is made.